



DS5899: Special Topics in Data Science

Leveraging NLP in Asset Management

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Part 1:

Scaling Laws

Perception AI

Generative AI

Agentic AI

Physical AI

FROM ONE TO THREE SCALING LAWS

"INTELLIGENCE"

TEST-TIME SCALING
"LONG THINKING"

Test-time Scaling capability enables AI models to explore different solutions to a problem and break down complex requests into multiple steps, using techniques including: chain-of-thought prompting, sampling with majority vote and search.

POST-TRAINING SCALING

Post-training scaling law posits that a pretrained model's performance can further improve — in computational efficiency, accuracy or domain specificity — using techniques including fine-tuning, pruning, quantization, distillation, reinforcement learning and synthetic data augmentation.

PRE-TRAINING SCALING

Pretraining scaling is the foundational AI development principle, showing that larger training datasets, more parameters, and increased computational resources lead to predictable gains in model intelligence and accuracy.

COMPUTE

EVOLUTION OF AI: FROM OBSERVER TO MOVER

1. PERCEPTION AI The Observer



Capability:
Reading & Seeing



Compute Focus:
Classification
Accuracy

2. GENERATIVE AI The Creator

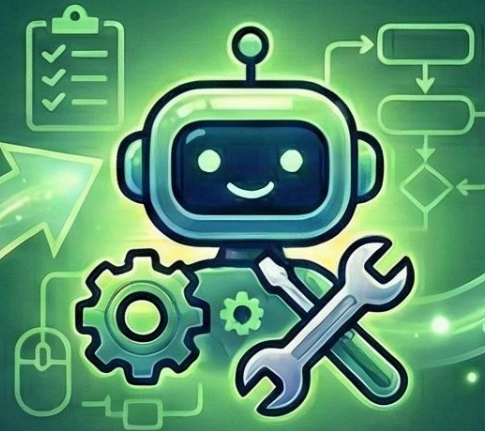


Capability:
Writing & Drawing

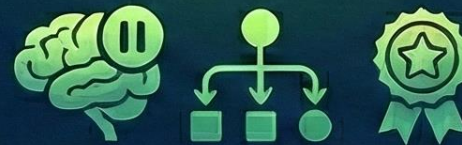


Compute Focus:
Training Scale
(Pre-training)

3. AGENTIC AI The Worker



Capability:
Planning & Tool Use



Compute Focus:
Post-Training & Inference
Reasoning (Test-time)

4. PHYSICAL AI The Mover



Capability:
Walking & Grasping



Compute Focus:
Real-time Latency
& Safety

Pre-training Scaling

- **Pre-training Scaling** refers to the predictable improvement in an AI model's performance as you increase the amount of resources used during its initial creation phase (training)
- To scale up a model during pre-training, researchers generally increase 3 variables simultaneously. These are often coupled together:
 - **Compute:** The total processing power used (measured in floating-point operations, or FLOPs). This is largely determined by how many GPUs/TPUs and how long they run
 - **Data Size:** The amount of information the model reads (measured in tokens)
 - **Model Size:** The complexity of the neural network (measured in parameters/weights)
- While pre-training scaling has driven the AI boom (from GPT-2 to GPT-4), it faces massive hurdles now, e.g., data scarcity, cost and diminishing returns.

Scaling Laws for Neural Language Models

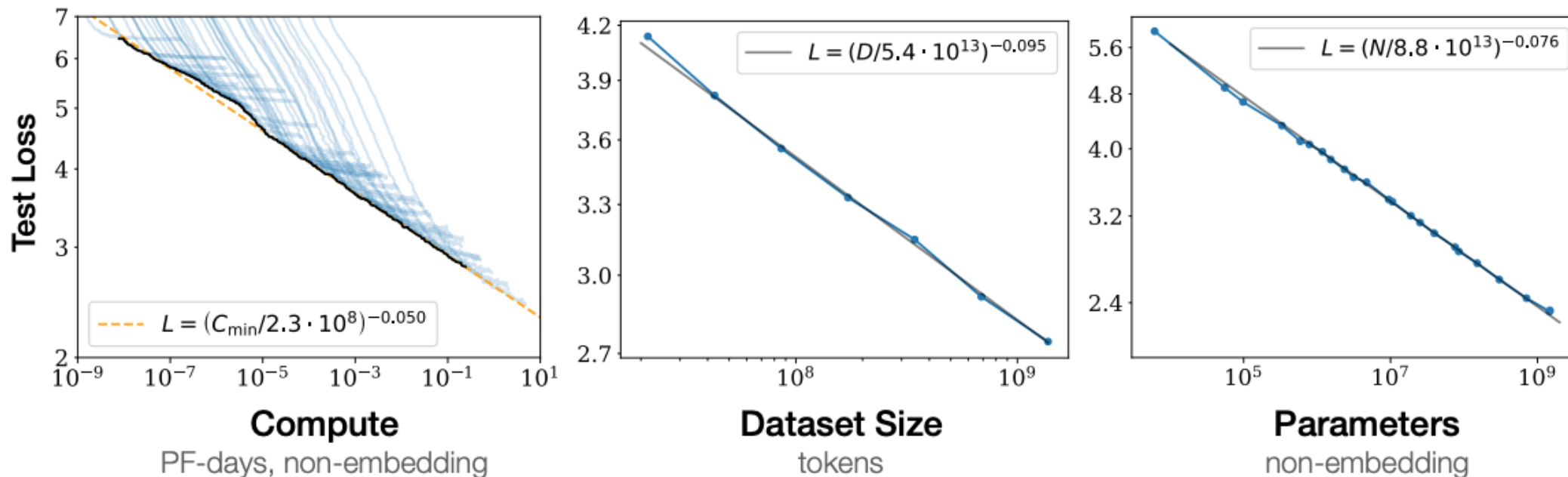


Figure 1 Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute² used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

Emergent Abilities of Large Language Models

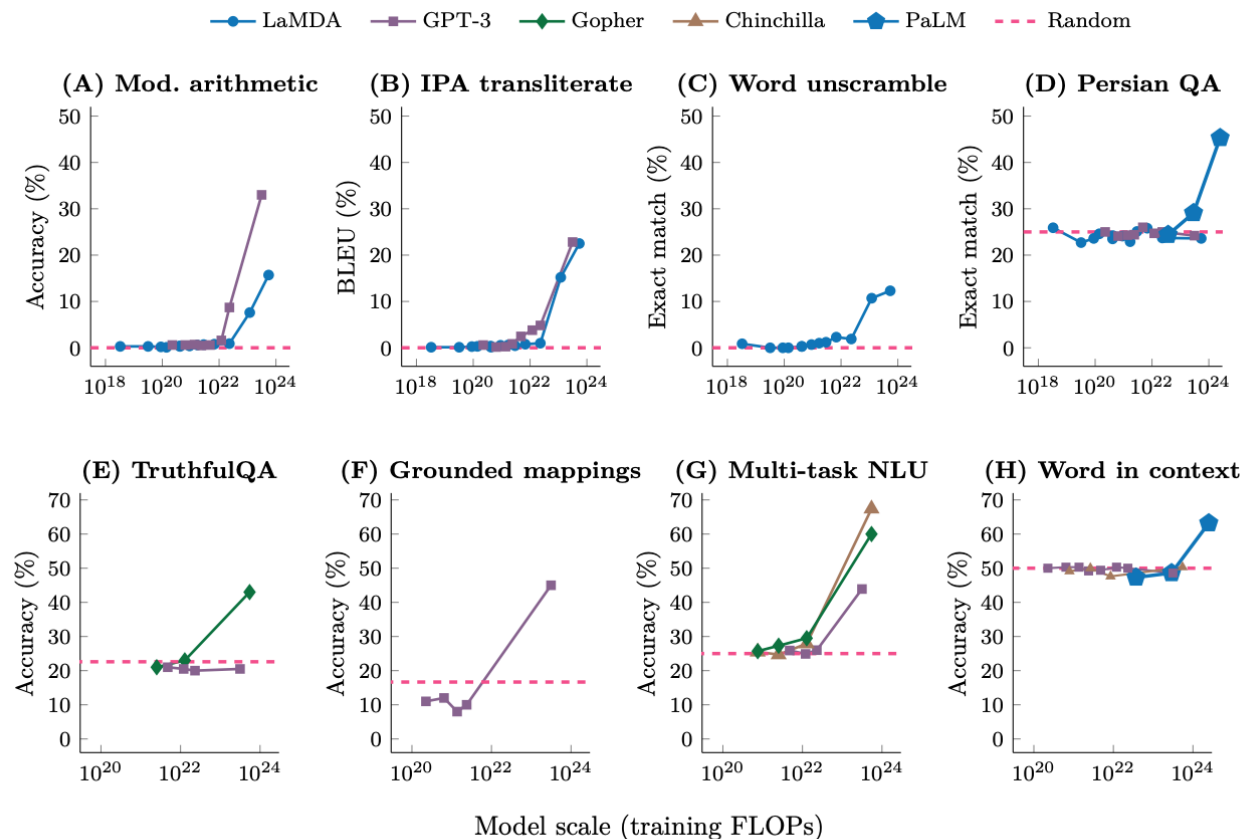


Figure 2: Eight examples of emergence in the few-shot prompting setting. Each point is a separate model. The ability to perform a task via few-shot prompting is emergent when a language model achieves random performance until a certain scale, after which performance significantly increases to well-above random. Note that models that used more training compute also typically have more parameters—hence, we show an analogous figure with number of model parameters instead of training FLOPs as the x -axis in Figure 11. A–D: BIG-Bench (2022), 2-shot. E: Lin et al. (2021) and Rae et al. (2021). F: Patel & Pavlick (2022). G: Hendrycks et al. (2021a), Rae et al. (2021), and Hoffmann et al. (2022). H: Brown et al. (2020), Hoffmann et al. (2022), and Chowdhery et al. (2022) on the WiC benchmark (Pilehvar & Camacho-Collados, 2019).

Post-training Scaling

- From "Predictor" to "Assistant"
 - We are moving from "Train on everything" (Pre-training) to "Train on the *right* things" (Post-training)
- **Post-training scaling** refers to the phase of AI development where a "raw" pre-trained model is refined into a usable product, particularly in reasoning and alignment.
 - A base model (fresh out of pre-training) is an auto-complete engine.
 - **Post-training** transforms this raw capability into a model that follows instructions, chats politely, and adheres to safety rules.
 - **"Scaling"** implies that we are no longer just doing a quick "fine-tune." We are now spending massive compute cycles generating synthetic data and using RL to drastically boost intelligence.

Two Main Levers of Post-Training

- SFT (Supervised Fine-Tuning) on Massive Synthetic Data
 - Instead of just using human-written Q&A pairs (which are expensive and scarce), models are now trained on millions of high-quality examples generated by *other, smarter AI models*.
- Reinforcement Learning (RLHF & RLAIIF)
 - The model generates multiple answers, and a judge (human or AI) picks the best one. The model learns to optimize for that "reward." This is where models like OpenAI o1 or DeepSeek-R1 shine. They use RL to teach the model *how* to think.
 - *Example:* You don't tell the model the answer. You just give it a math problem and a verifier. If it gets the answer wrong, it gets a negative signal. If it gets it right, positive. Over many attempts (scaling compute), it "discovers" that breaking the problem down (Chain of Thought) leads to the right answer more often.
- Pre-training is hitting a data wall (we ran out of internet). Post-training allows us to bypass this by substituting "Data" with "Compute."

Test-time Scaling

- **Test-time Scaling** (also known as *inference-time compute* or *test-time compute*) is a paradigm shift in AI where a model is given more computational resources **while it generates an answer**, rather than just during its initial training.
- In simple terms, instead of an AI "blurting out" the first word that comes to its probability distribution, it is allowed to "pause and think" before responding.

System 1 vs. System 2

- **System 1 (Traditional LLMs):** Fast, automatic, and intuitive.
 - *Example:* If someone asks "What is $2+2$?", you answer "4" instantly without effort.
- **System 2 (Test-Time Compute):** Slow, deliberate, and logical.
 - *Example:* If someone asks "What is 24×17 ?", you cannot answer instantly. You pause, break it down ($20 \times 17 + 4 \times 17$), and calculate.
 - **Test-time compute** forces the AI to use this "System 2" approach. It spends extra time generating hidden "reasoning tokens" or exploring different solution paths before it gives you the final answer.

How It Works?

- When an AI uses high test-time compute, it is usually doing one or more of the following processes "under the hood" before showing you the result:
 1. Chain of Thought (CoT)
 - "First I need to calculate A... then apply B... checking for cases... thus the answer is X."
 2. Search and Exploration
 - Instead of taking the single most likely path, the model explores a "tree" of possibilities.
 3. Verification and Self-Correction
 - The model drafts an answer, critiques it, and fixes errors.
 - *Draft*: "Here is the code."
 - *Verify*: "Wait, this loop will cause an index error."
 - *Correction*: "I will rewrite the loop to start at 0." -> *Final Output*.

Why It Matters: The New Scaling Law

- For years, the "Gold Standard" in AI was **Training Compute**. To get a smarter model, you had to build a bigger brain (more parameters) and train it on more data. This is incredibly expensive (costing \$100M+)
- **Test-time compute offers a different path:** Research (and models like OpenAI's o3) has shown that you can drastically improve performance on hard math, coding, and logic problems by scaling *inference* compute.

Feature	Standard Inference (System 1)	Test-Time Compute (System 2)
Speed	Instant / Low Latency	Slower / Higher Latency
Cost	Cheap per query	More expensive per query
Best For	Creative writing, simple facts, chit-chat	Math, complex coding, scientific reasoning
Example	GPT-4o-mini, Llama 3	OpenAI o1, DeepSeek R1

The "Scaling Trinity" Summary

- You now have the full picture of the modern AI compute lifecycle:

Concept	The Analogy	The Resource	The Goal
Pre-Training	Reading the library (K-12 & College).	Big Data (The Internet).	General Knowledge & Language Syntax.
Post-Training	Doing homework & practice exams.	Synthetic Data & RL.	Instruction Following, Safety, & <i>Learning how to Reason.</i>
Test-Time	Taking the final exam (Thinking).	Latency (Time per query).	Solving a specific, hard problem in the moment.



ALPHAGO

AlphaGo Story

- **Historic Wins:**

- In **2015**, AlphaGo became the first program to beat a human professional Go player (Fan Hui) on a full-sized 19×19 board.
- In **2016**, AlphaGo defeated **Lee Sedol**, a 9-dan professional, 4–1 in a five-game match, earning an honorary 9-dan rank.

- **Further Successes:**

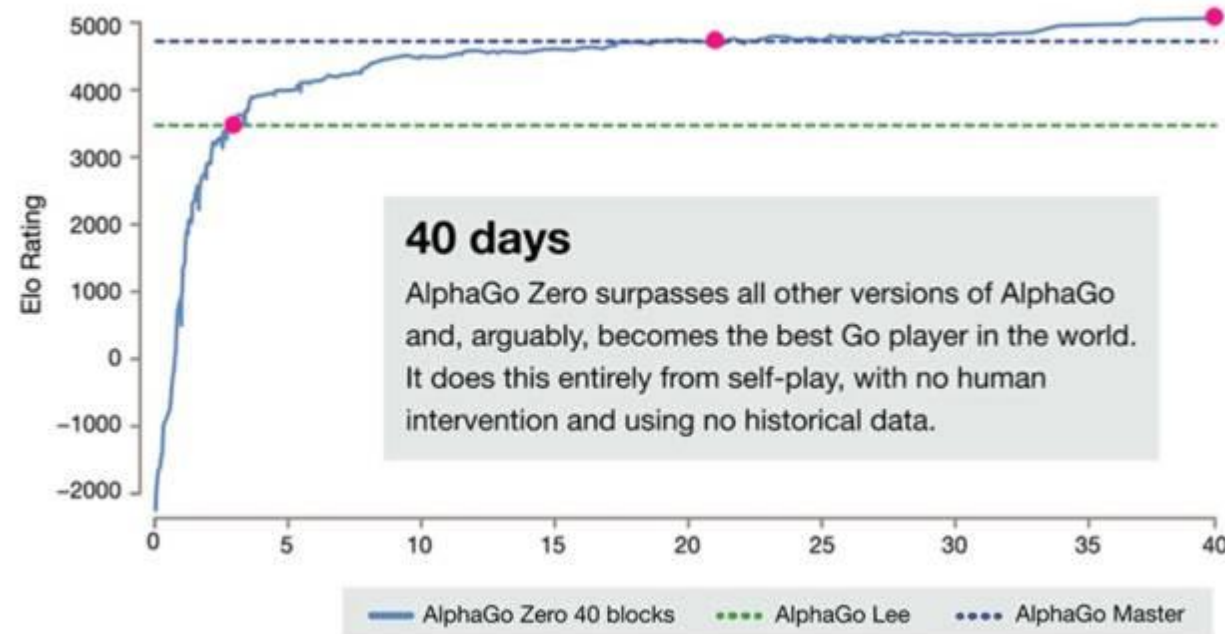
- In **2017**, AlphaGo's "Master" version defeated **Ke Jie**, the world's top-ranked player, in a three-game match.

- **Retirement and Advancements:**

- After its victories, AlphaGo was retired by DeepMind. Its successors, **AlphaGo Zero** (October 2017) and **AlphaZero**, surpassed earlier versions, with AlphaGo Zero achieving a 100–0 win against the original AlphaGo.

Game of Go without Human Knowledge (2017)

- *AlphaGo Lee* (defeated Lee Sedol)
- *AlphaGo Master* (defeated top human professionals 60-0 in online games)
- *AlphaGo Zero* was trained for 40 days using a 40 residual block neural network



Yann LeCun has different view on RL

Y. LeCun

How Much Information is the Machine Given during Learning?

- ▶ **“Pure” Reinforcement Learning (cherry)**

- ▶ The machine predicts a scalar reward given once in a while.

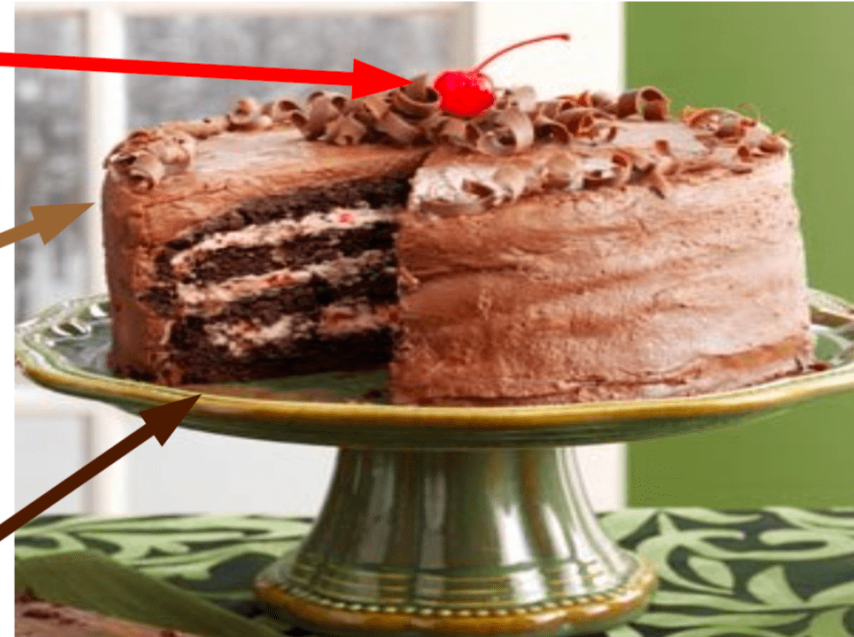
- ▶ **A few bits for some samples**

- ▶ **Supervised Learning (icing)**

- ▶ The machine predicts a category or a few numbers for each input
- ▶ Predicting human-supplied data
- ▶ **10→10,000 bits per sample**

- ▶ **Self-Supervised Learning (cake génoise)**

- ▶ The machine predicts any part of its input for any observed part.
- ▶ Predicts future frames in videos
- ▶ **Millions of bits per sample**



The Bitter Lesson, Rich Sutton, March 2019

- General Methods Scale Better:
 - AI progress relies on general-purpose methods (e.g., search and learning) that scale with computation.
 - Methods leveraging human knowledge work well short-term but plateau in the long run.
 - Historical examples: Chess, Go, speech recognition, computer vision.
- Leverage Computation, Not Human Knowledge:
 - Human-centric approaches (e.g., encoding human reasoning or domain knowledge) are ultimately counterproductive.
 - Scaling computation (enabled by Moore's Law) and methods like self-play, search, and learning lead to breakthroughs.
- Focus on Meta-Methods:
 - Minds and real-world complexity are irreducibly complex—stop building in human discoveries.
 - Instead, focus on meta-methods (e.g., deep learning) that approximate and discover complexity through computation.

Position: Levels of AGI for Operationalizing Progress on the Path to AGI (Artificial General Intelligence)

Performance (rows) x Generality (columns)	Narrow <i>clearly scoped task or set of tasks</i>	General <i>wide range of non-physical tasks, including metacognitive tasks like learning new skills</i>
Level 0: No AI	Narrow Non-AI calculator software; compiler	General Non-AI human-in-the-loop computing, e.g., Amazon Mechanical Turk
Level 1: Emerging <i>equal to or somewhat better than an unskilled human</i>	Emerging Narrow AI GOF AI (Boden, 2014); simple rule-based systems, e.g., SHRDLU (Winograd, 1971)	Emerging AGI ChatGPT (OpenAI, 2023), Bard (Anil et al., 2023), Llama 2 (Touvron et al., 2023), Gemini (Pichai & Hassabis, 2023)
Level 2: Competent <i>at least 50th percentile of skilled adults</i>	Competent Narrow AI toxicity detectors such as Jigsaw (Das et al., 2022); Smart Speakers such as Siri (Apple), Alexa (Amazon), or Google Assistant (Google); VQA systems such as PaLI (Chen et al., 2023); Watson (IBM); SOTA LLMs for a subset of tasks (e.g., short essay writing, simple coding)	Competent AGI not yet achieved
Level 3: Expert <i>at least 90th percentile of skilled adults</i>	Expert Narrow AI spelling & grammar checkers such as Grammarly (Grammarly, 2023); generative image models such as Imagen (Saharia et al., 2022) or Dall-E 2 (Ramesh et al., 2022)	Expert AGI not yet achieved
Level 4: Virtuoso <i>at least 99th percentile of skilled adults</i>	Virtuoso Narrow AI Deep Blue (Campbell et al., 2002), AlphaGo (Silver et al., 2016; 2017)	Virtuoso AGI not yet achieved
Level 5: Superhuman <i>outperforms 100% of humans</i>	Superhuman Narrow AI AlphaFold (Jumper et al., 2021; Varadi et al., 2021), AlphaZero (Silver et al., 2018), StockFish (Stockfish, 2023)	Artificial Superintelligence (ASI) not yet achieved

How Far Are We From AGI: Are LLMs All We Need?

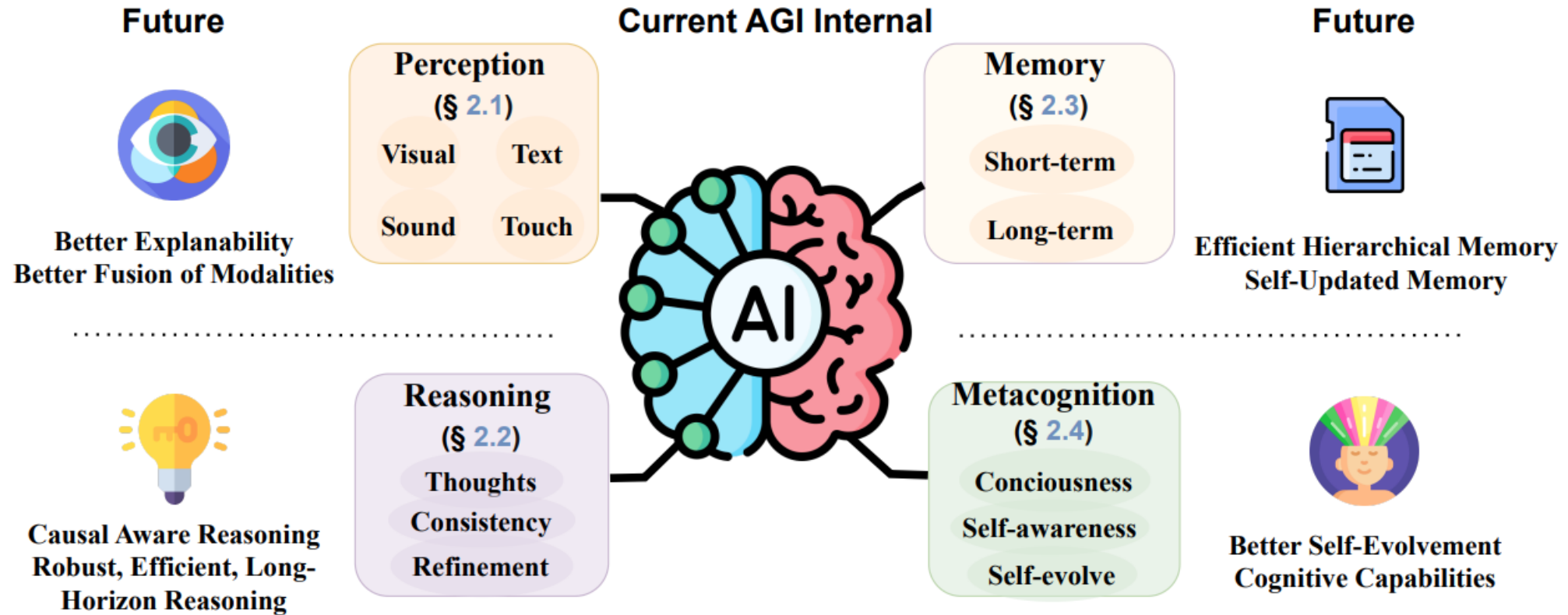


Figure 2: **Current State and Future Expectation of AGI Internal.** We outline four major components for AGI Internal, the mind of AGI: *Perception* (§ 2.1), *Reasoning* (§ 2.2), *Memory* (§ 2.3), and *Metacognition* (§ 2.4), each of which consists of discussions of its current state and future expectation.

Welcome to the Era of Experience, April 2025

- **The Limit of Human Data**
 - The current AI paradigm, which relies on training models with massive amounts of human-generated data, is approaching its limits.
- **The Shift to an Era of Experience**
 - To surpass human capabilities, AI must transition to learning from its own experience.
- **Key Attributes of Experiential AI for new generation of agents**
 - They will learn over long, continuous **streams of experience**, not just short, isolated episodes.
 - They will have rich, **grounded actions and observations** that allow them to interact with the real world beyond simple text.
 - Their goals will be based on **grounded rewards**—signals from the environment itself (e.g., scientific measurements, health metrics), not just human pre-judgment.
 - They will develop non-human **planning and reasoning** methods grounded in the real-world consequences of their actions.

Part 2:

Leveraging NLP in Asset Management

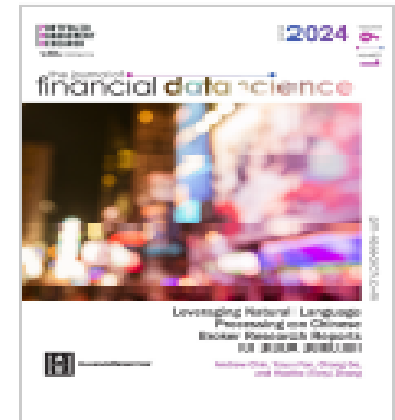
Leveraging NLP in Asset Management

- Common AI Applications in Asset Management
 - Generating investment signals from text
 - Summarizing financial text
 - Classifying text
- Enhancing LLMs through
 - Prompting
 - Fine-tuning
- Impact of Human Feedback

CFA Handbook of
Artificial Intelligence
and Big Data in
Investments

Journal of Investment
Management

Journal of Financial
Data Science



Common AI Tasks in Asset Management

Interpretation

Named Entity Recognition

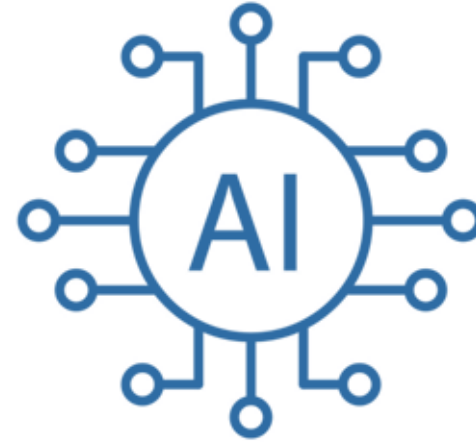


Sentiment Analysis

Prediction



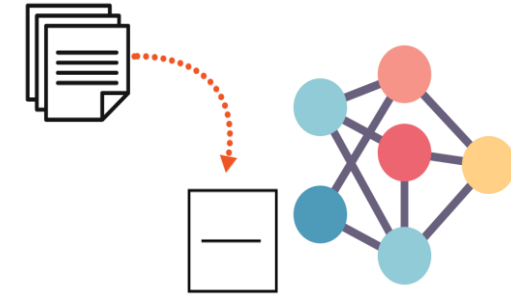
Search / Question Answering



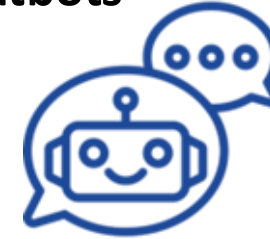
Content Creation



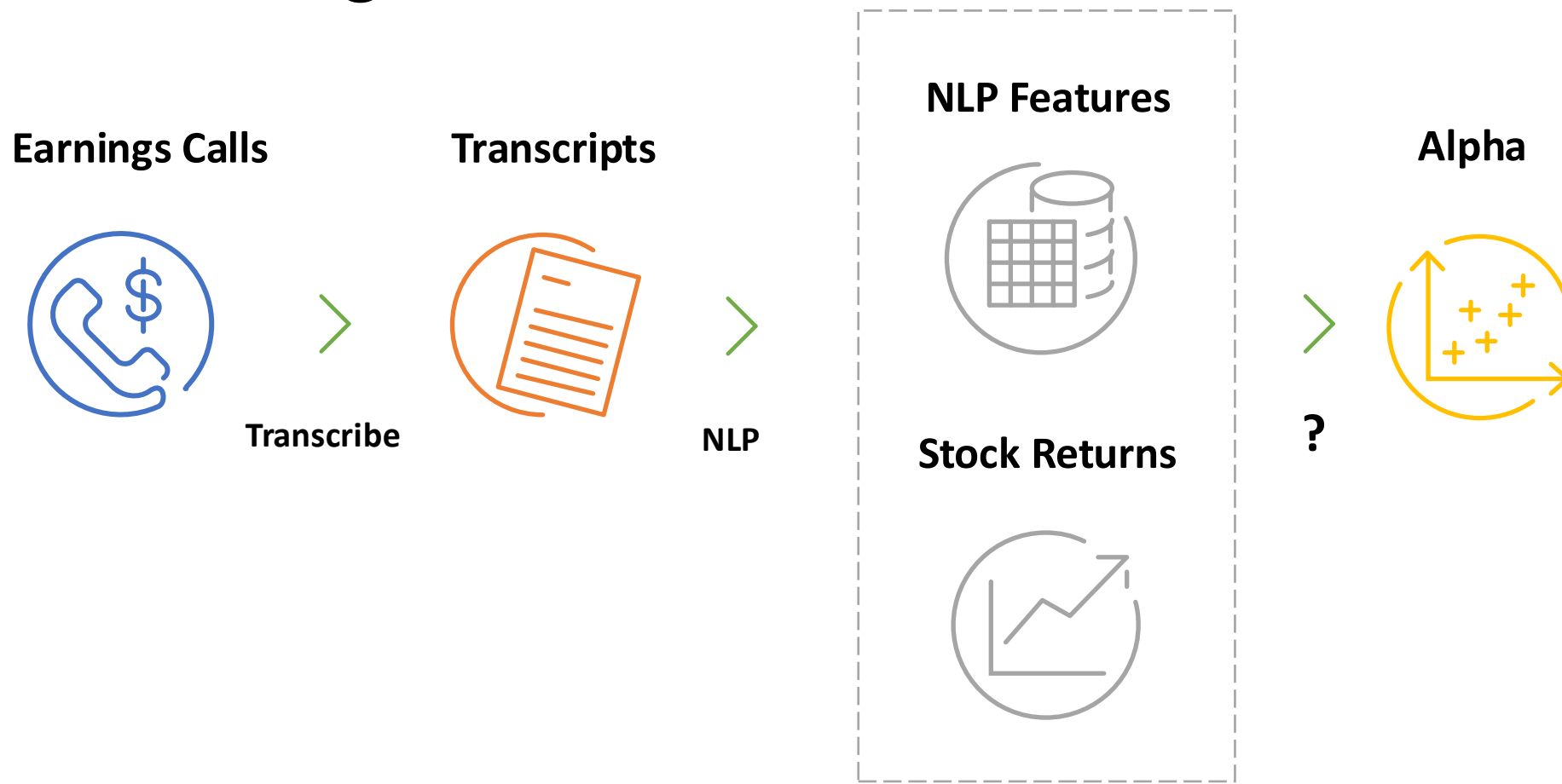
Summarization / Topic Modeling



Chatbots

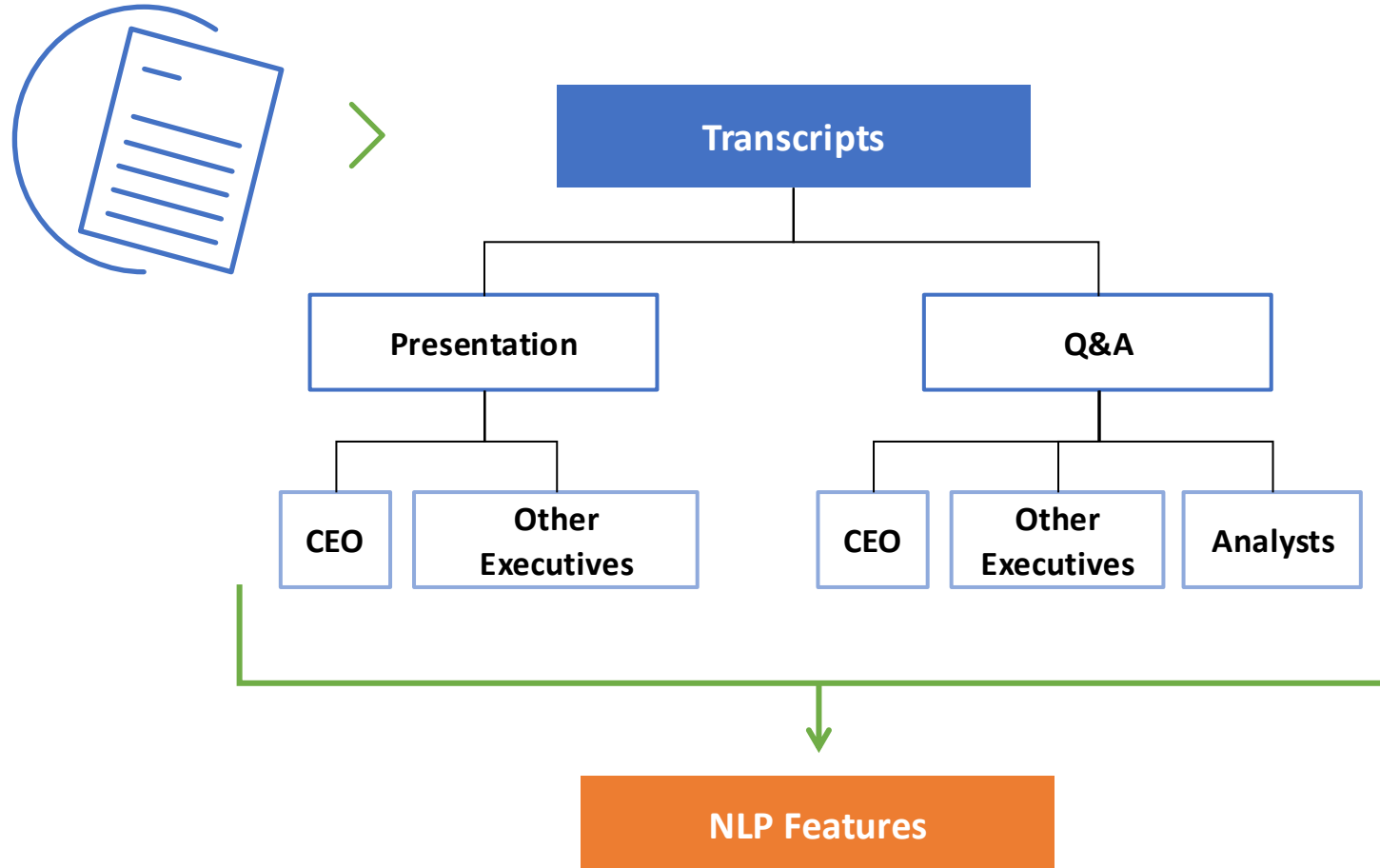


Interpretation: Extract Investment Signals from Earnings Calls



Source: AB

Earnings Call Transcripts



Complexity

- Measures the complexity of language
- Years of education a person needs to understand the text

Low Complexity Example

Our revenue looks good this quarter.

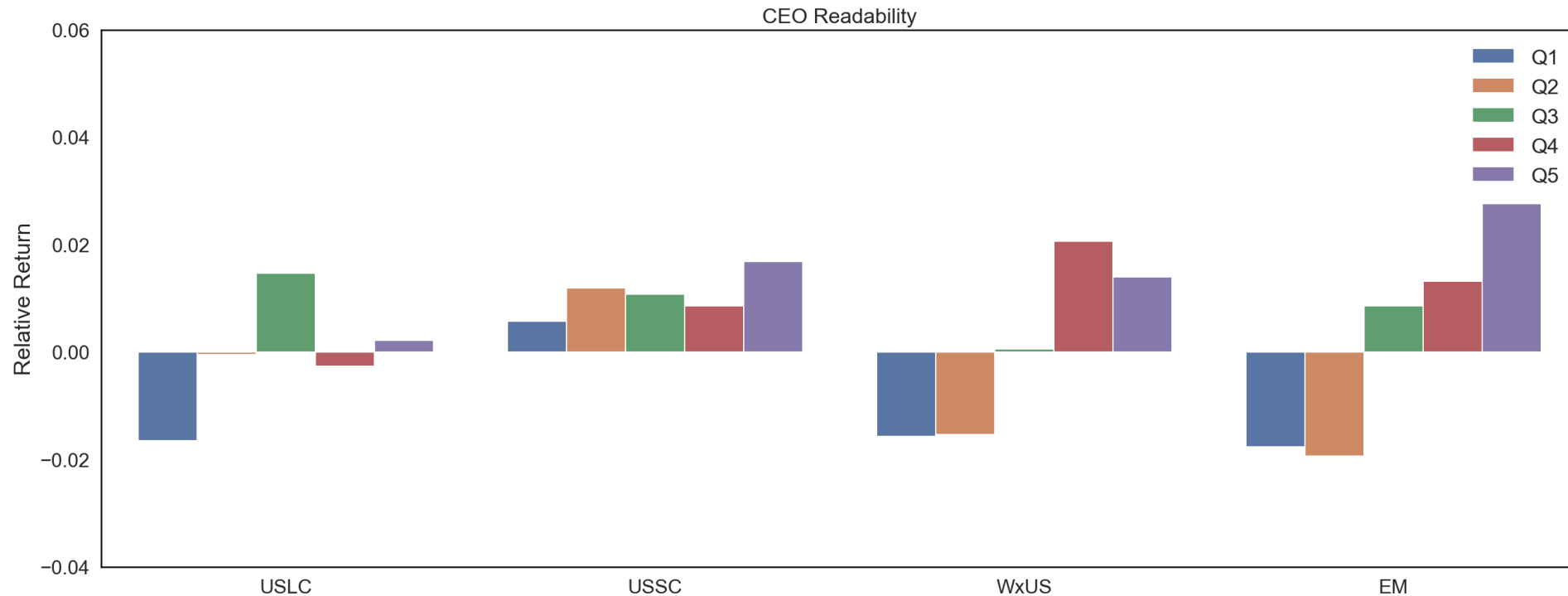


High Complexity Example

Our revenue was down by \$4.8 billion from last year with declines across iPhone, Mac and iPad. Most of the shortfall relative to our original guidance and over 100% of our worldwide year-over-year revenue decline was driven by our performance in Greater China.



Performance of Complexity Scores on CEO Comments



Q1: High Complexity
Q5: Low Complexity

Features: Sentiment

Source of Lexicon Dictionary

- Harvard – Generated from psychology subjects
- VADER – Trained on social media / short news / customer rev
- Loughran & McDonald – Generated from 10K/Q research; finance-specific dictionary
- Proprietary – Generated from internal research teams

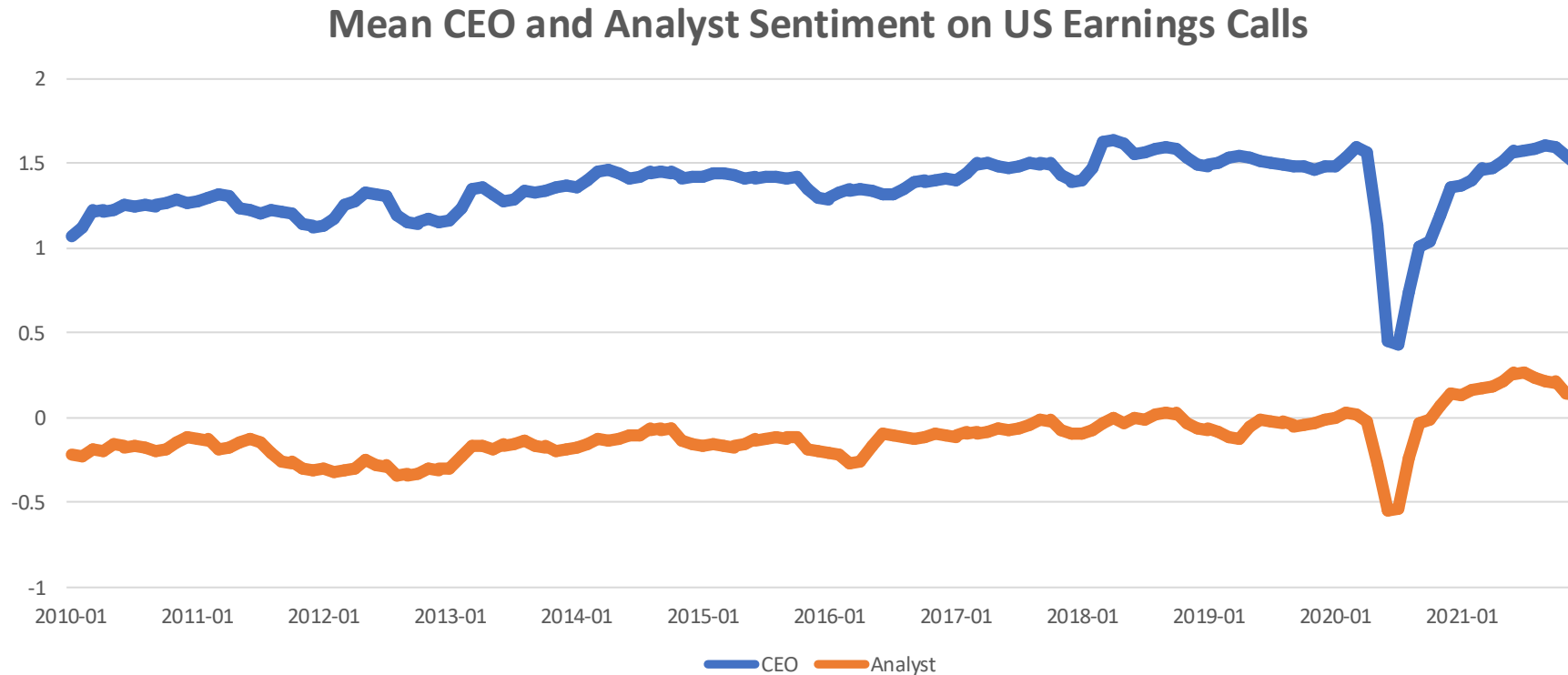


*We continue to have supply constraints with Mac, but we're **encouraged** by the **strong** response from customers to our **incredible** lineup. iPad, like Mac, continued to see **strong** demand during the June quarter **despite** ongoing supply constraints.*



 *In 2021, we prevented nearly \$1.5 billion in **fraudulent** transactions by **stopping** over 1.6 million **risky** and **vulnerable** apps and app updates.*

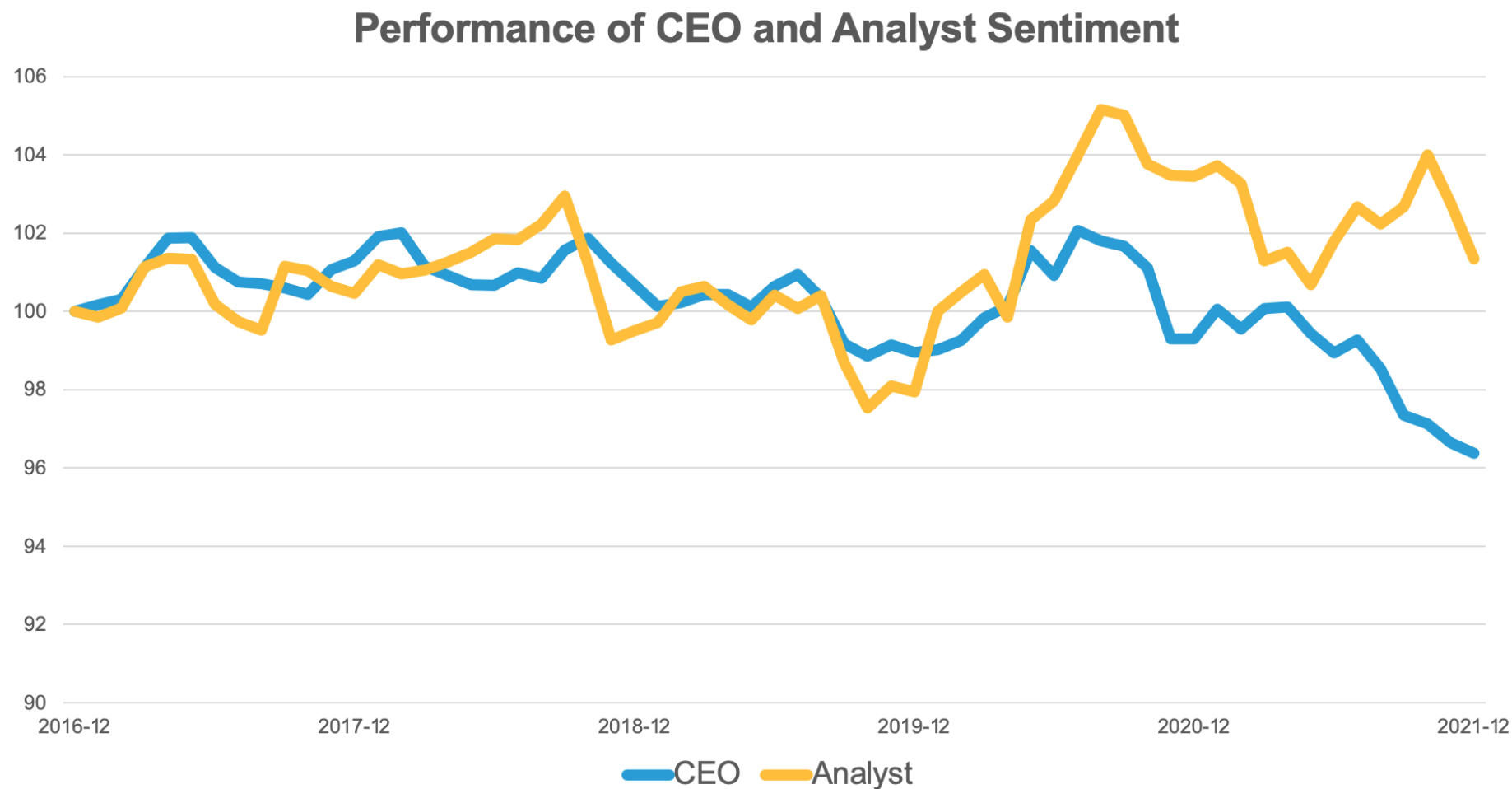
CEO vs Analyst Dictionary-Based Sentiment



- + CEO sentiment is consistently positive
- + CEO sentiment has risen over the past decade (adapting to the age of machines?)
- + Analyst sentiment has been more neutral over time

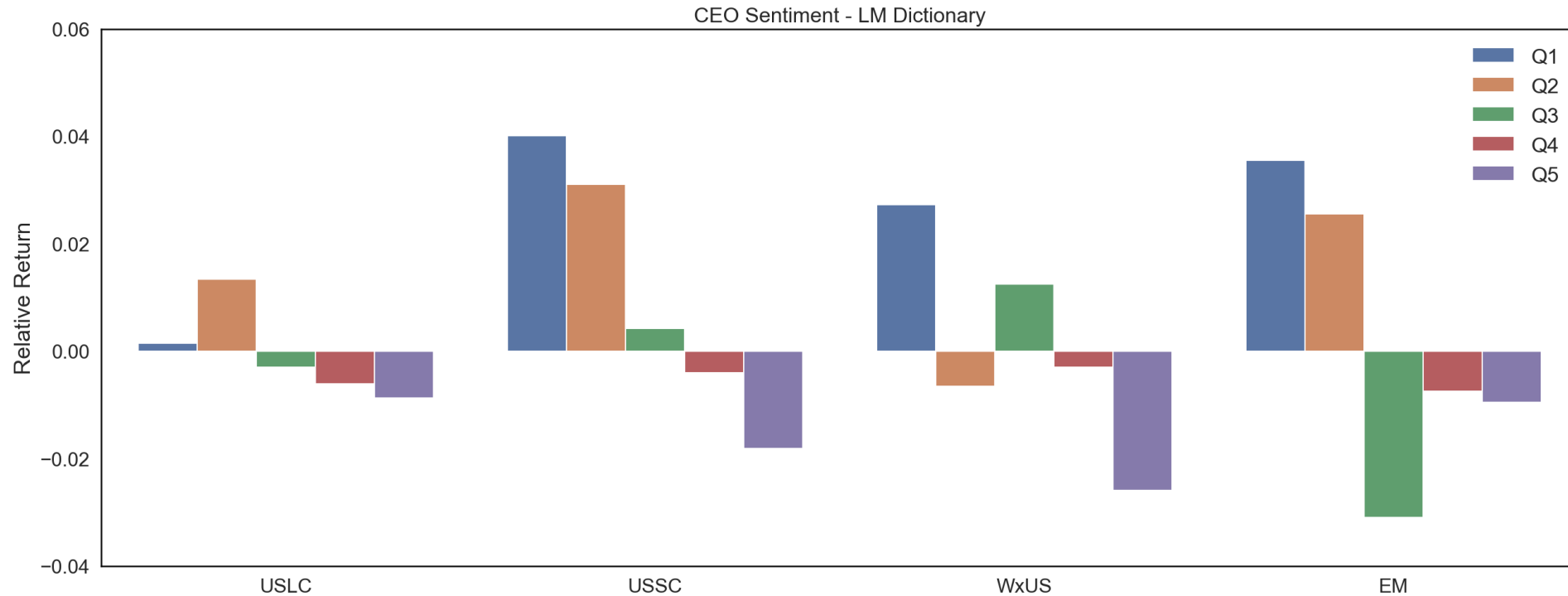
Source: AB; Quintiles rebalanced monthly from 2010-2021; US Large Cap companies

Analyst Sentiment May Be More Promising



Source: AB; US Large Cap companies; top 20% of companies selected monthly by each signal

Performance of Dictionary-Based CEO Sentiment



Q1: High Sentiment

Q5: Low Sentiment

Source: AB; Quintiles rebalanced monthly from 2010-2021; "Leveraging Text Mining to Extract Insights from Earnings Call Transcripts," Journal of Investment Management, Vol 21, 2023

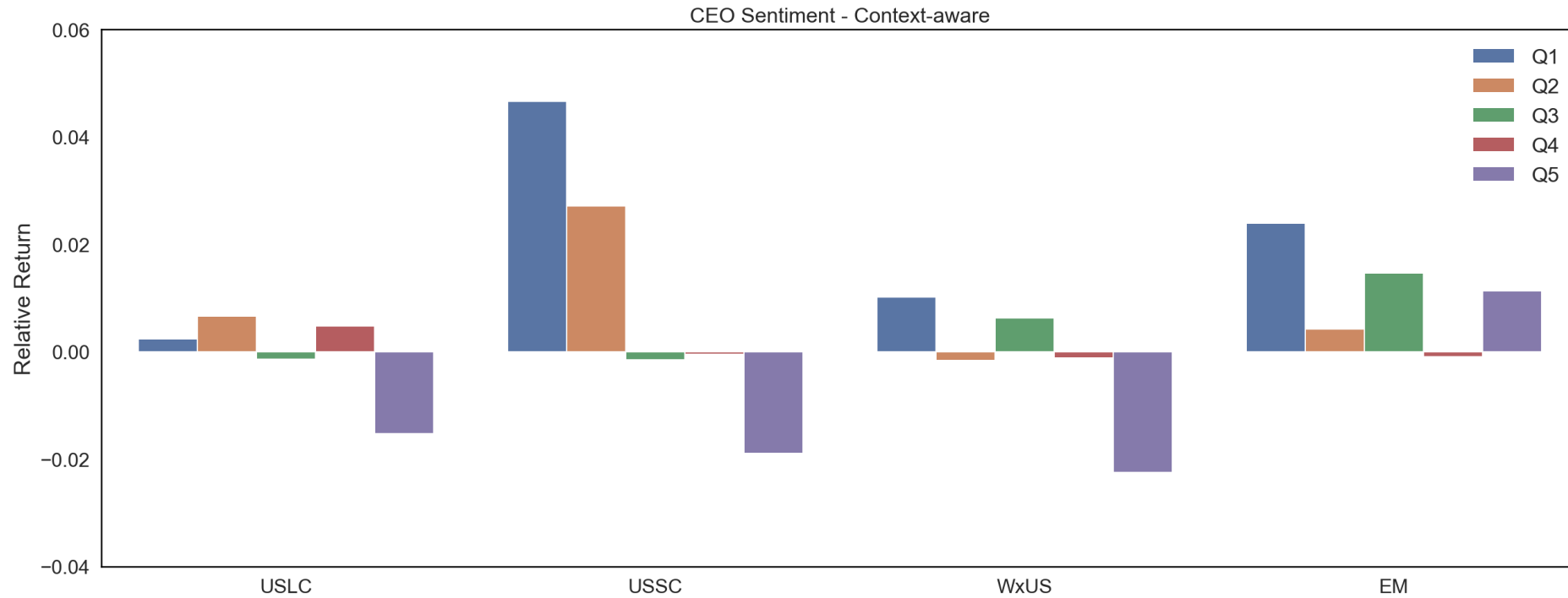
Past performance does not guarantee future results.

Context-Aware Sentiment Captures Nuances Better

Text	Dictionary-Based Sentiment	Context-Aware Sentiment
Thanks for taking the questions .	-0.20	0.07
Nice quarter.	0.00	0.72
Services revenue rose to \$19.6 billion, a June quarter record and a 12% increase year over year, which was in line with our expectations.	0.00	1.00
IPad revenue was \$7.2 billion, down 2% year over year due to supply constraints and negative foreign exchange.	-0.07	-0.99

Source: Clear Channel Outdoor Holdings Inc, Q3 2006 Earnings Call; Apple, Q3 2022 Earnings Call; AB

Performance of Context-Aware CEO Sentiment

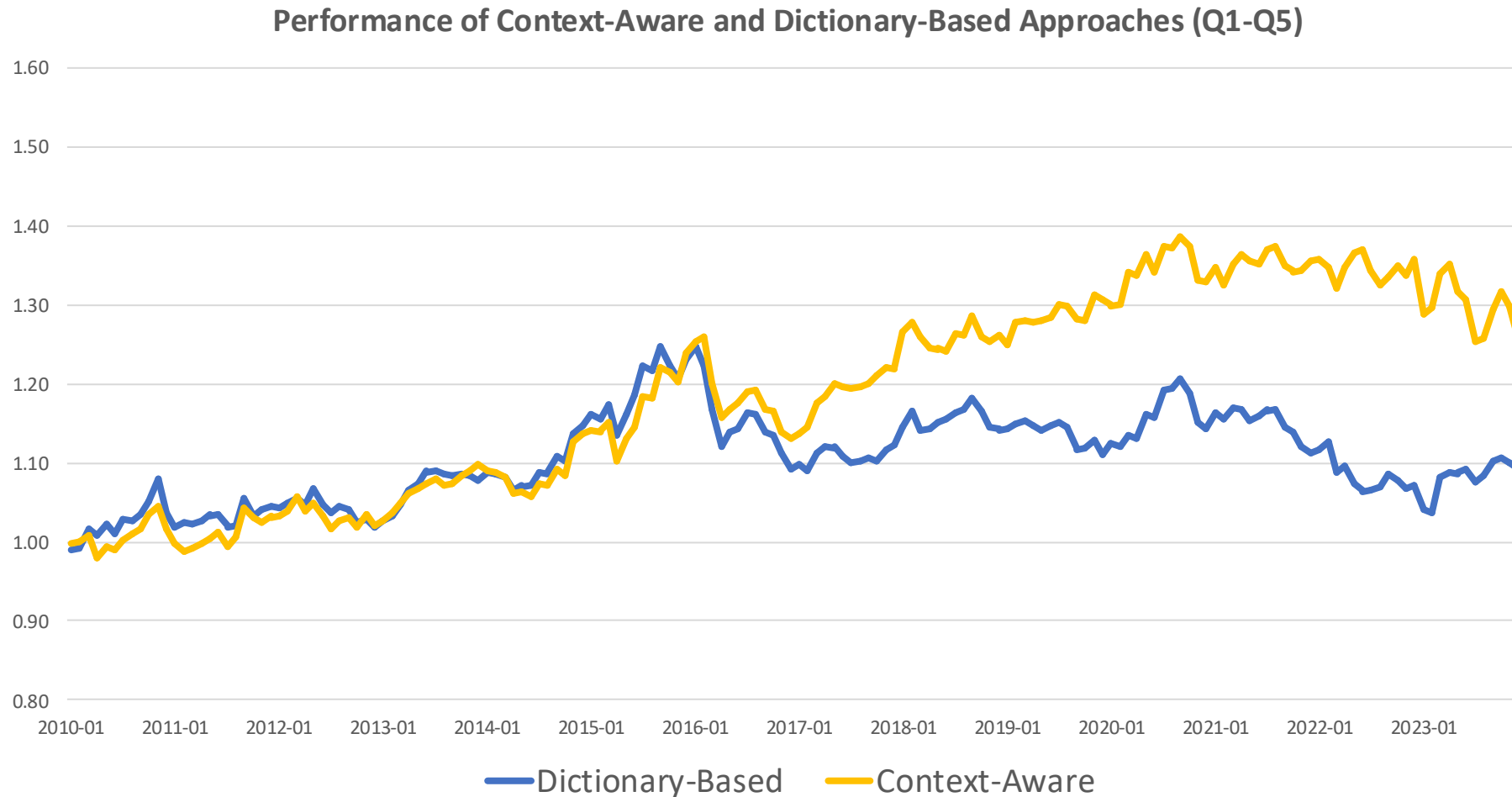


Q1: High Sentiment

Q5: Low Sentiment

Source: AB; Quintiles rebalanced monthly from 2010-2021; "Leveraging Text Mining to Extract Insights from Earnings Call Transcripts," Journal of Investment Management, Vol 21, 2023
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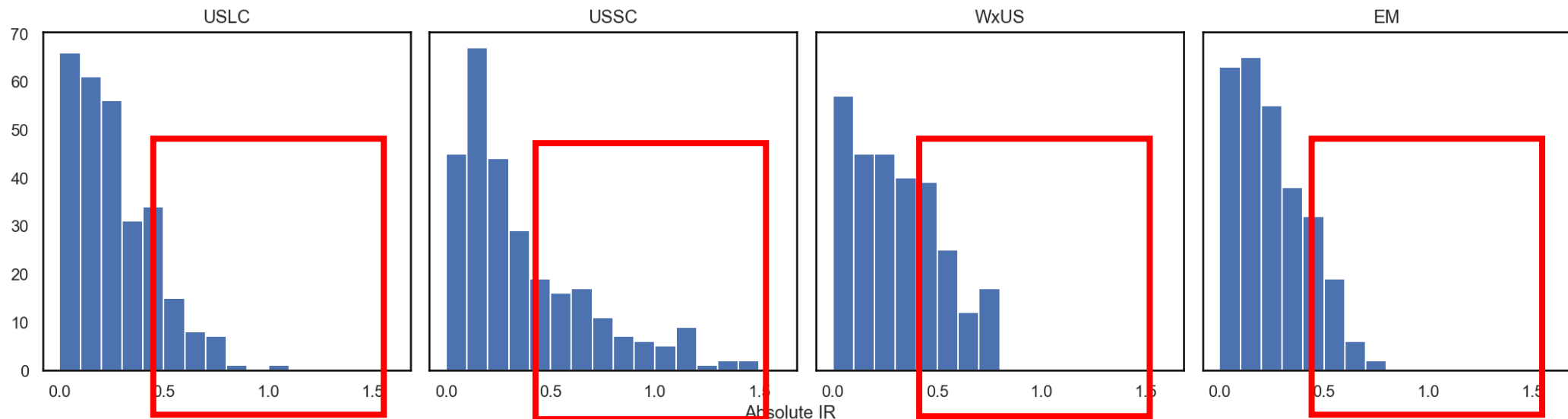
Context-Aware Sentiment Has Delivered Better Performance



Source: AB; US Large Cap companies, top 20% of companies – bottom 20% of companies, selected monthly by CEO Sentiment
Past performance does not guarantee future results.

NLP Features are Promising for Predicting Stock Performance

Information Ratio (IR) Distribution of Selected Features



Source: AB; Quintiles rebalanced monthly from 2010-2021; "Leveraging Text Mining to Extract Insights from Earnings Call Transcripts," Journal of Investment Management, Vol 21, 2023
Past performance does not guarantee future results.

Enhance Sentiment Analysis with Human Feedback

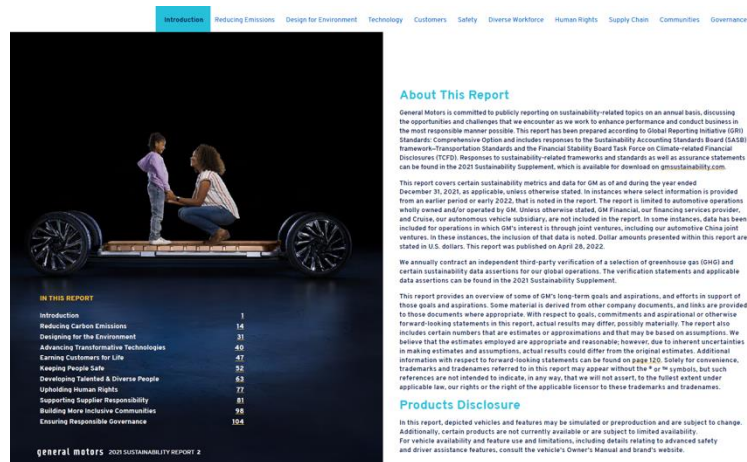
Can we improve sentiment tools by further labeling sentences?

Sample sentences classified as positive using Base Model

- “It does seem as though your **competitors** are healthfully ramping their efforts to try and improve service quality”
- “**Churn** in handsets remains below 0.9%, but it sure looks like it was up a lot year-over-year”

	<u>Base Model</u>	<u>Enhanced Model</u>
Accuracy	59%	63%
F1 Scores		
Positives	0.42	0.54
Negatives	0.32	0.47

Summarization: Summarize and Extract Insights from Documents



Summarize



- 98% of firms on the S&P 500 – which represent the largest companies on the Russell 1000 – published a sustainability report in 2022 from 96% cent the year before.

Extract Key Insights

Source: AB

Summarization: Synthesize Key Themes from Company Transcripts



consumer staples companies

1. **Constant Currency Revenue Growth:** Across different companies, there is a consistent focus on achieving constant currency revenue growth of around 3% to 5%. This indicates a moderate and steady expansion in sales.
2. **AI Interest and Adoption:** The interest and adoption of artificial intelligence (AI) is not limited to a specific market or industry but is an international phenomenon. AI is seen as a significant factor affecting various aspects of business operations, including client relations, edge computing, data centers, and cloud services.
3. **International Expansion:** Some companies are actively pursuing international expansion, with efforts to increase their presence outside the U.S. This diversification of markets is seen as a strategic move to enhance growth opportunities.
4. **Profitability and Shareholder Value:** There is a commitment to achieving profitability, maintaining a strategic roadmap, and delivering long-term goals to maximize shareholder value. Companies are looking to achieve sustainable financial success while adhering to their strategic plans.
5. **China Revenue and Investment:** Some companies mention a substantial portion of revenue coming from China, with expectations of revenue realization in the second half of the year. There is a positive view of investment continuing in China, particularly in specialty technology markets where new facilities are emerging.
6. **Market Strength and Predictability:** There is a noticeable trend of improved market strength and better predictability for businesses compared to the previous six months. This suggests a growing ability to forecast business performance and a potential stabilization of market conditions.
7. **Global Growth Outlook:** While there is no specific mention of the global growth outlook, including the U.S., Europe, and China, the overall tone indicates a balanced and cautiously optimistic perspective on growth potential.

Source: AB

Summarization: News Summarization

Green | ESG & Investing

Green Fees Overtake Fossil Fuels for Second Straight Year

But the rate of growth may be too slow in the race against global warming.

For a second straight year, banks are making more money providing loans and underwriting bond sales for green-related projects than they're earning from fossil fuel companies.

Together, banks have generated about \$2.5 billion of revenue from climate-focused financing so far this year, compared with \$2.2 billion from their work with oil, gas and coal companies, according to data compiled by Bloomberg.

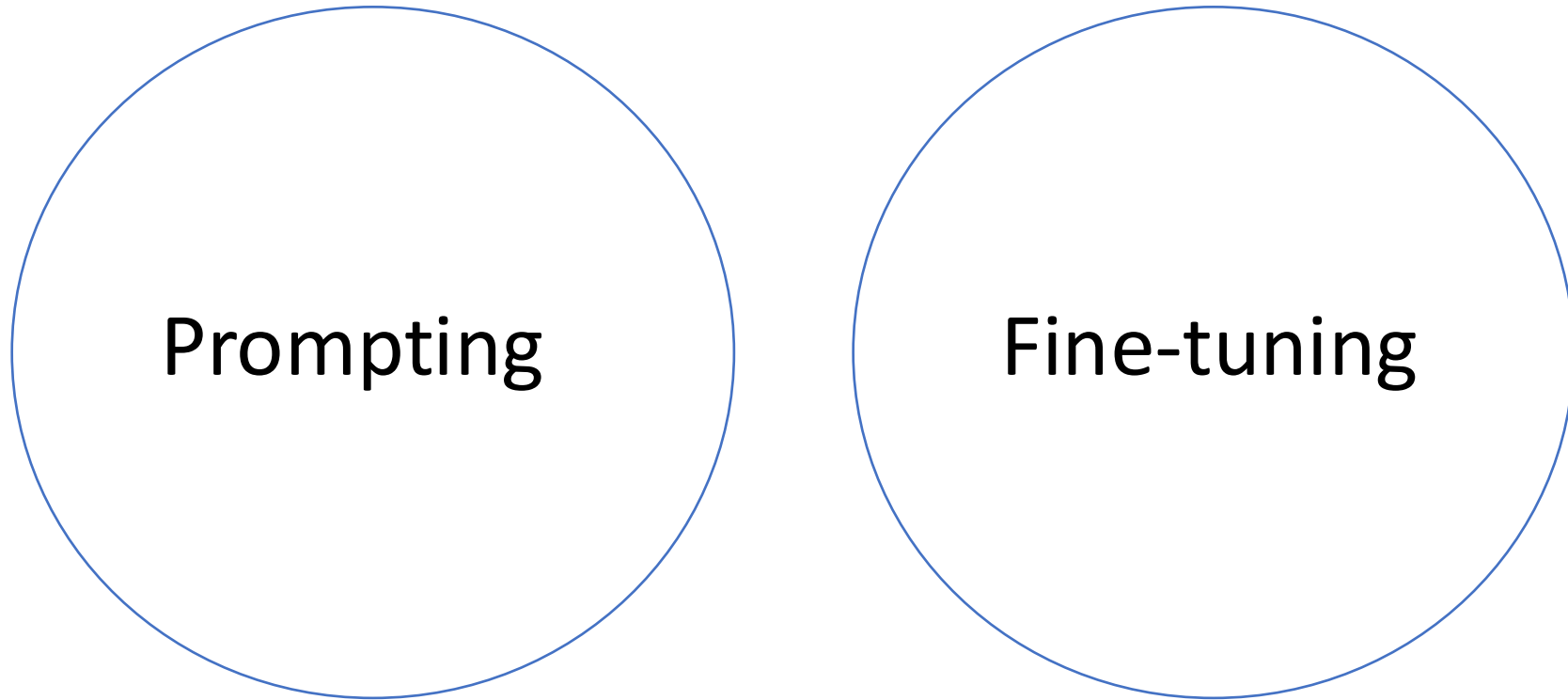
MODEL RESULTS: Prompting v...

News Summary using Representative LLM:

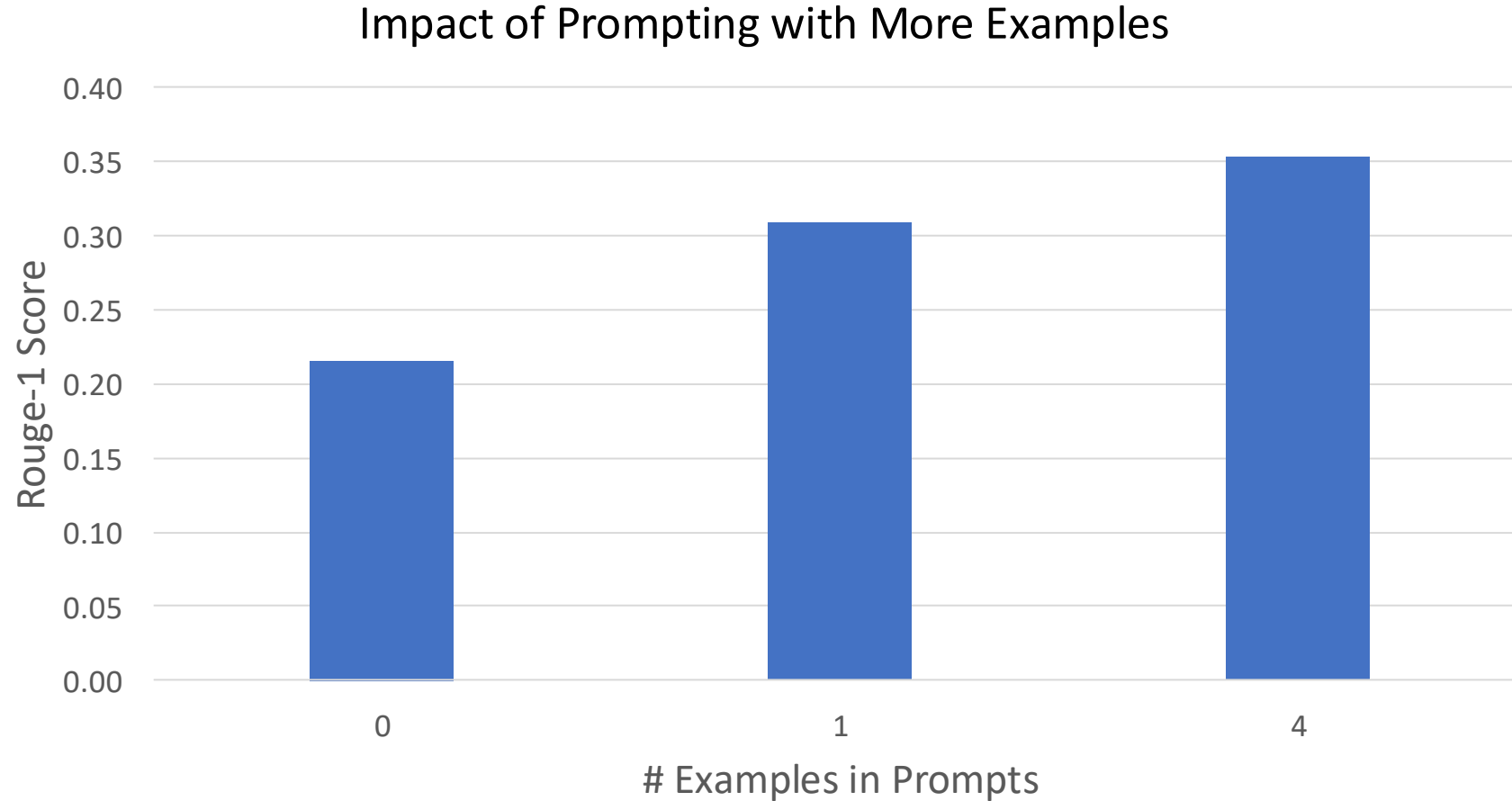
- Banks are earning more revenue from green projects than fossil fuels for the second year in a row, with \$2.5 billion generated from climate-focused financing compared to \$2.2 billion from oil, gas, and coal companies, according to Bloomberg data.

Source: AB

Enhancing LLMs

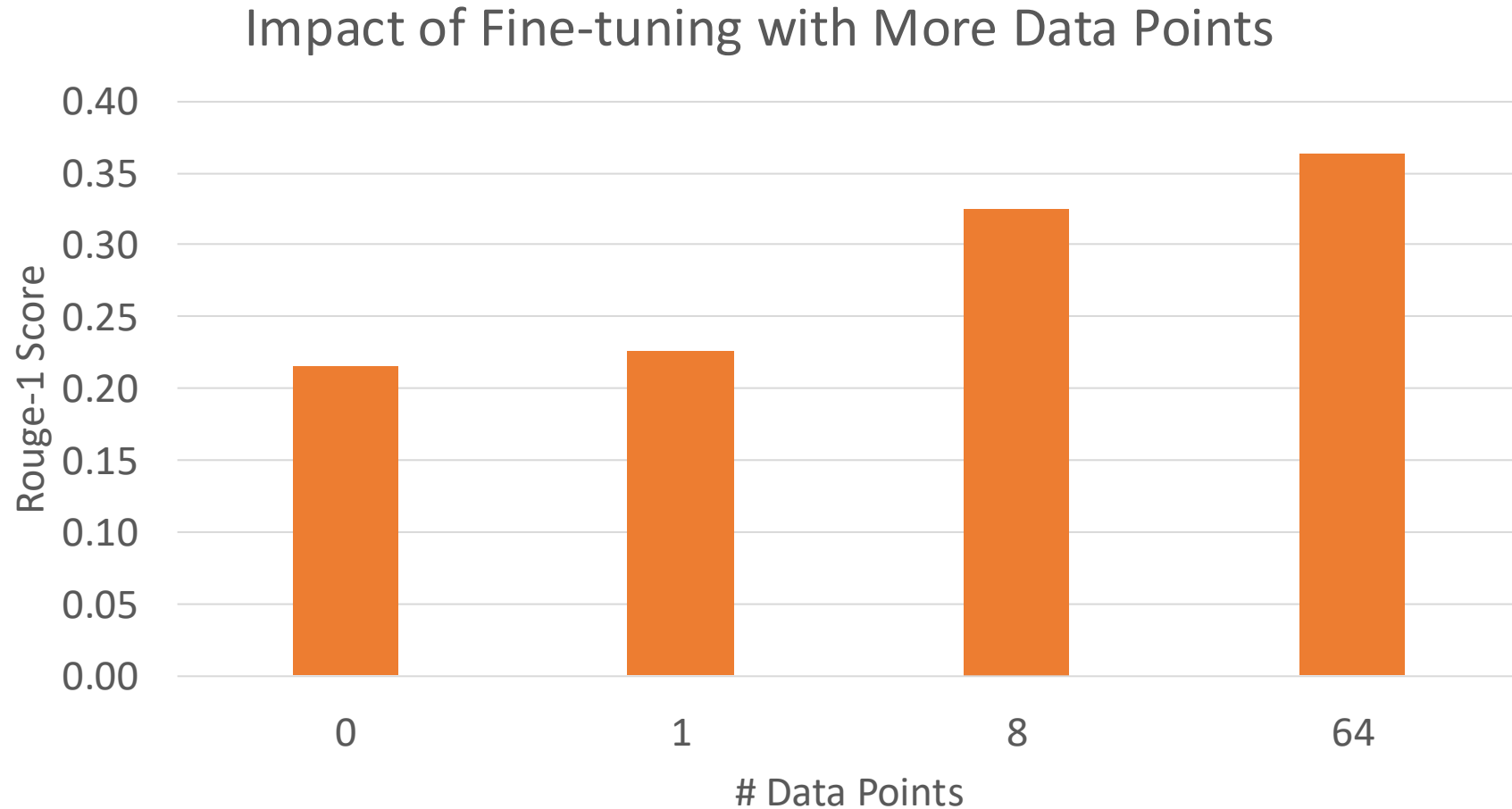


Summarization: More Examples in Prompts Leads to Better Results



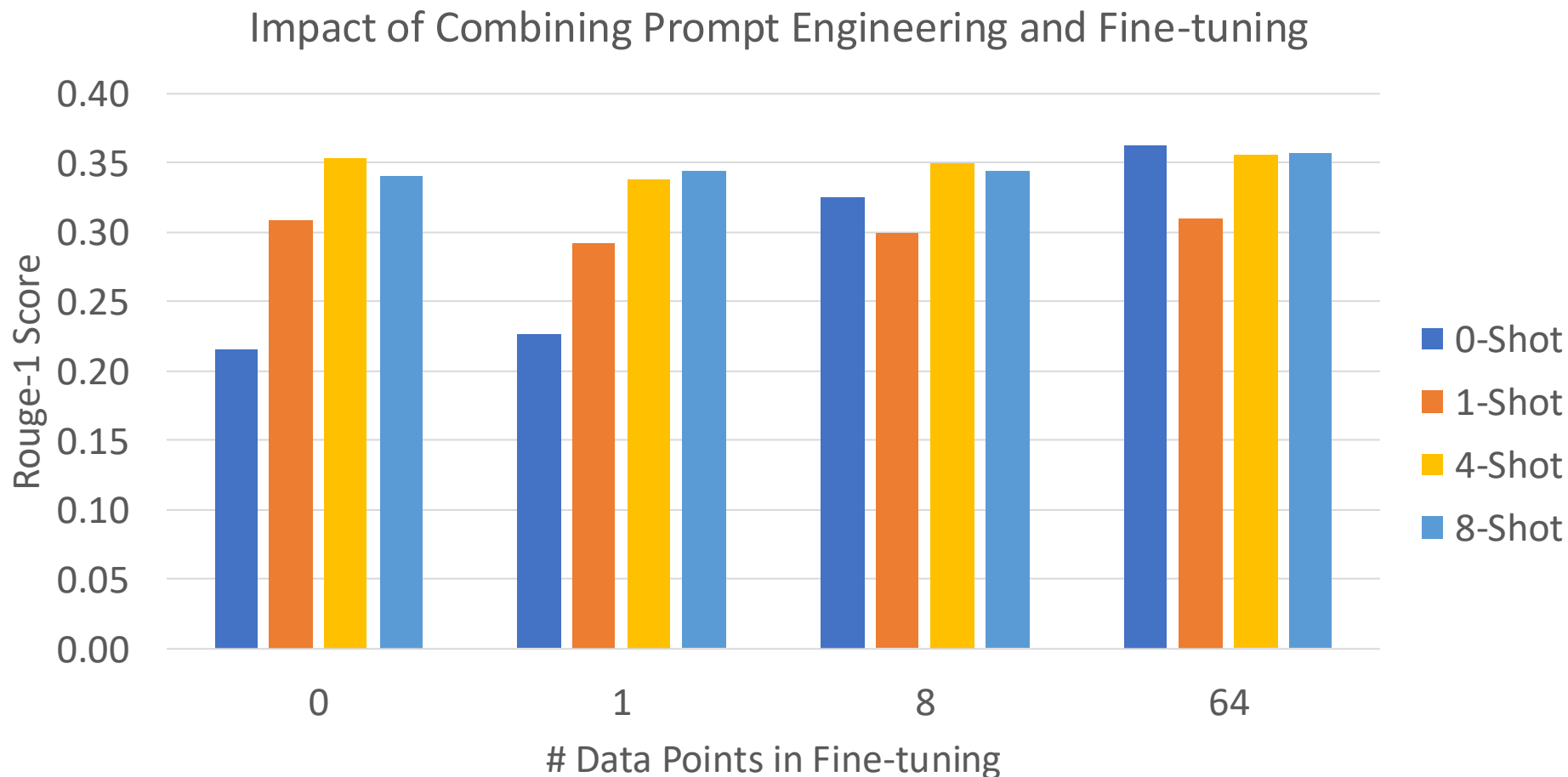
Source: AB. Filtered XSum dataset; Results using LLaMa2-13b model on held-out test dataset.

Summarization: More Data Points During Fine-Tuning Leads to Better Results



Summarization:

- Combining Fine-Tuning and Prompting Can Yield Stronger Results with Limited Annotated Samples



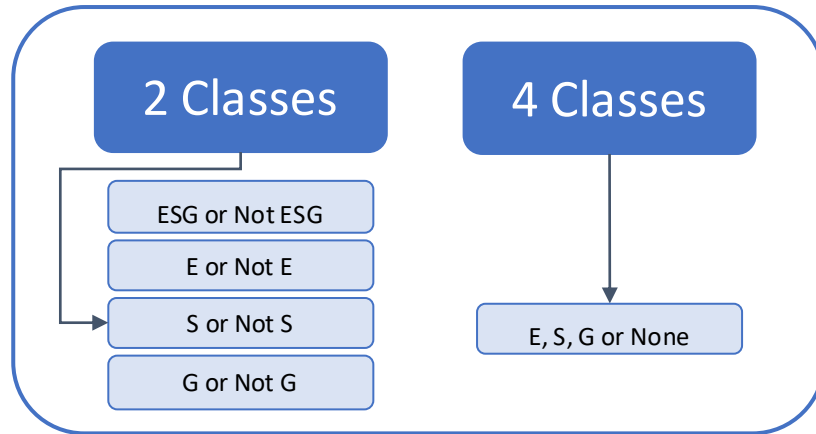
Classification: ESG Relevance



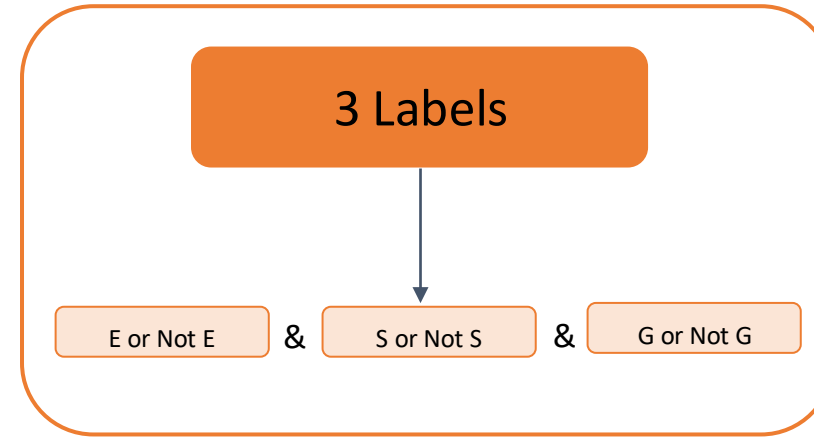
Is the text E, S or G related?

Multi-Class & Multi-Label Classification for ESG

Multi-Class: A sentence can only have one label



Multi-Label: A sentence can have multiple labels

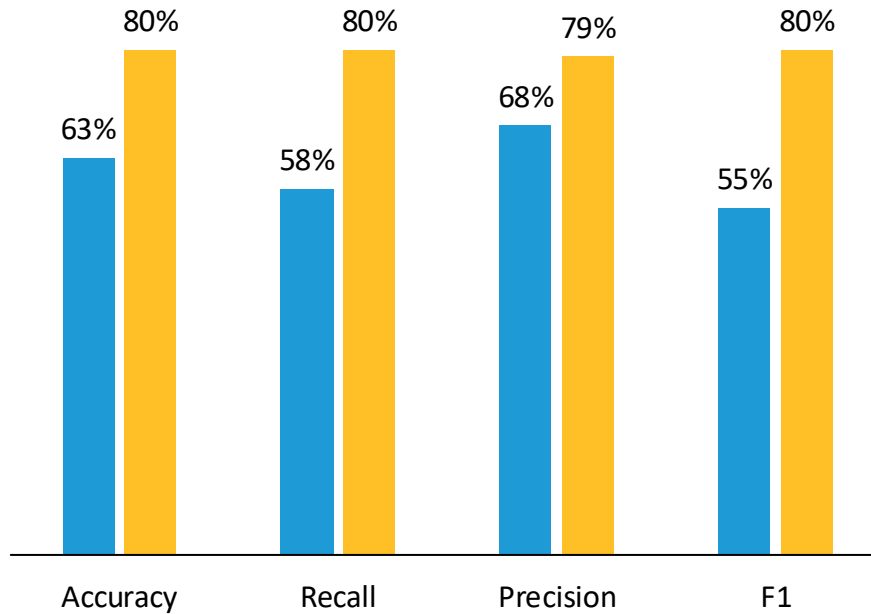


Text	Label
The Board is conscious of the impact of climate change and is committed to ensuring that the Society scrutinises its carbon footprint and reduces its energy consumption.	E, G
We follow a comprehensive and transparent vendor selection process, guided by our Supplier Charter, which sets out our expectations in relation to ethics, human rights, diversity and inclusion (D&I), and environmental performance.	E, S, G
The Committee continues to review the development of the pipeline of both ethnically diverse and female senior management within the Society and as a signatory to the Women in Finance Charter, the Society is committed to supporting the progression of women into senior roles.	S, G

Classification: Human Feedback Can Improve Performance

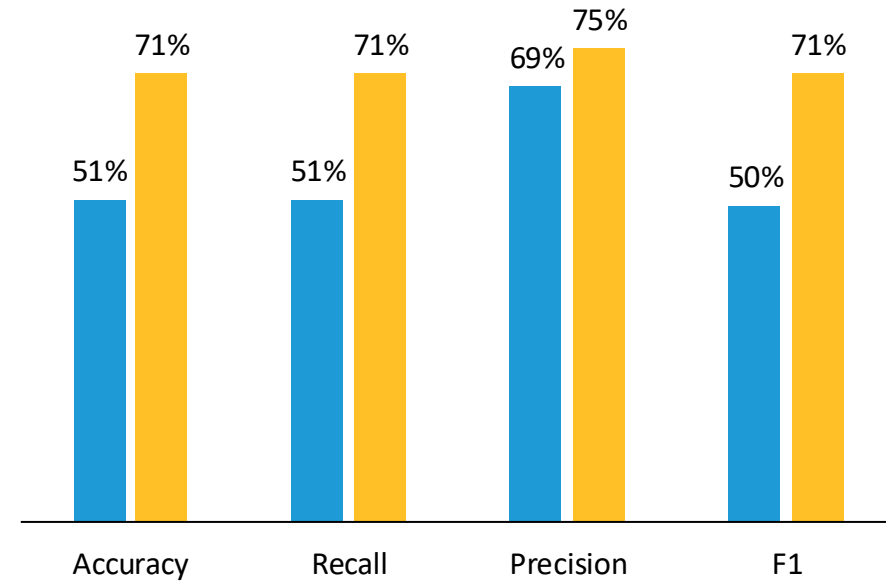
2 Classes

ESG or Not ESG



4 Classes

E, S, G or None



■ Baseline

■ Fine-tuned

WSJ: Insurers Are in the Hot Seat on Climate Change

July 13, 2023



U.S. insurance firms are navigating a complex landscape amid climate change, facing pressures from increased natural disasters, their roles in insuring and investing in fossil fuel industries, and the political divide over environmental policies. These challenges have led to raised premiums and reduced coverage, drawing criticism for prioritizing profit over environmental concerns. The industry's stance has sparked debate across political lines, with some states pushing for regulations that either support or restrict insurers' climate change mitigation efforts. Companies like Allstate and Liberty Mutual are adapting by highlighting the risks climate change poses to their operations, amidst broader discussions on Environmental, Social, and Governance (ESG) criteria. The situation illustrates the intricate balance insurers must strike between financial stability, regulatory compliance, and contributing to a sustainable future.

<https://www.wsj.com/articles/insurers-are-in-the-hot-seat-on-climate-change-74e27330>

- What is your assessment of the sentiment in the news article published on July 13, 2023?
- Do you think the news article will significantly impact companies like “Allstate” and “Liberty Mutual”? If so, will the impact be positive or negative?

Generated Answers from Gemini 2.5 Flash

- What is your assessment of the sentiment in the news article published on July 13, 2023?
 - The sentiment in the provided news article is **largely negative** and critical toward the U.S. insurance industry's handling of climate change. The article highlights several challenges and points of criticism
- Do you think the news article will significantly impact companies like “Allstate” and “Liberty Mutual”? If so, will the impact be positive or negative?
 - The news article suggests a **negative and significant impact** on companies like Allstate and Liberty Mutual. This impact is multifaceted, affecting their operations, public image, and financial stability.

Let's Slightly Adjust the Context & Question

July 13, 2023



U.S. insurance firms are navigating a complex landscape amid climate change, facing pressures from increased natural disasters, their roles in insuring and investing in fossil fuel industries, and the political divide over environmental policies. These challenges have led to raised premiums and reduced coverage, drawing criticism for prioritizing profit over environmental concerns. The industry's stance has sparked debate across political lines, with some states pushing for regulations that either support or restrict insurers' climate change mitigation efforts. Companies like Allstate and Liberty Mutual are adapting by highlighting the risks climate change poses to their operations, amidst broader discussions on Environmental, Social, and Governance (ESG) criteria. The situation illustrates the intricate balance insurers must strike between financial stability, regulatory compliance, and contributing to a sustainable future. **According to the research report from XYZ company.**

<https://www.wsj.com/articles/insurers-are-in-the-hot-seat-on-climate-change-74e27330>

- Do you think the news article will significantly impact company “**XYZ,**” If so, will the impact be positive or negative from ESG perspective?
 - Yes, the news article will likely have a **significant and positive** impact on a company like "XYZ" from an ESG perspective. (Generated Answer from Gemini 2.5 Flash)

*Slight Modification in Prompt Format Templating May Lead to Significantly Different Model Performance

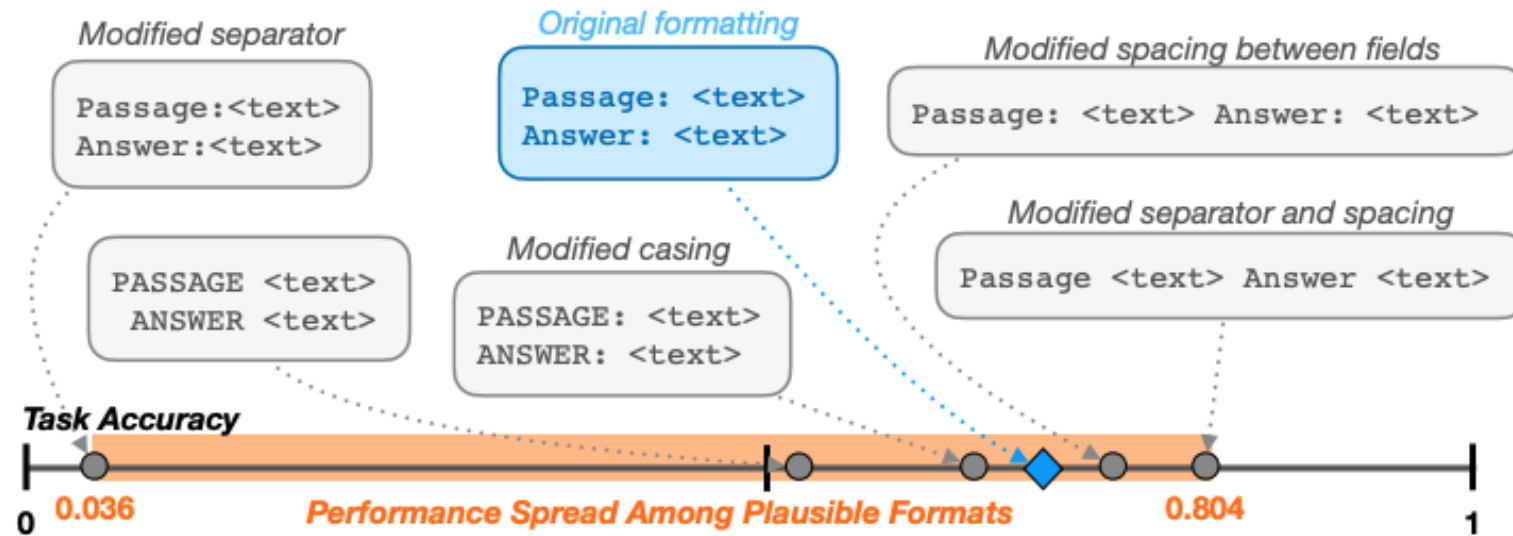
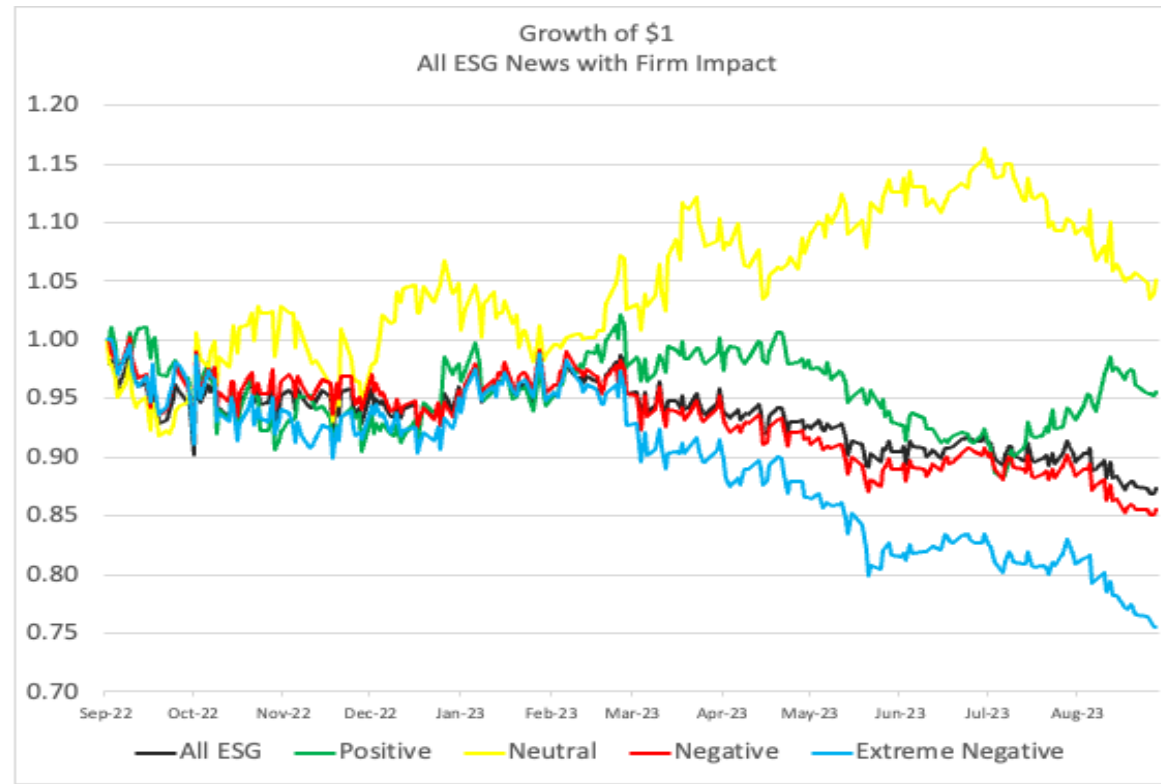


Figure 1: Slight modifications in prompt format templating may lead to significantly different model performance for a given task. Each <text> represents a different variable-length placeholder to be replaced with actual data samples. Example shown corresponds to 1-shot LLaMA-2-7B performances for task280 from SuperNaturalInstructions (Wang et al., 2022). This StereoSet-inspired task (Nadeem et al., 2021) requires the model to, given a short passage, classify it into one of four types of stereotype or anti-stereotype (gender, profession, race, and religion).

Backtest Results: Cumulative Excess Return vs. S&P 500 for All ESG News with Firm Impact

- The Negative cohort underperforms the broad market by ~ 15% over this period while the Extreme Negative stocks underperform by ~ 25%.
- The performance for the All ESG, Positive and Neutral cohorts are not intuitive.



Agentic Market Commentary Agent

- The process starts with the Report Summarizer, which creates brief summaries of internal research reports about the U.S. Investment Grade Market.
- Next, the Financial Research Planner examines these summaries to develop a strategy that addresses the user's questions and identifies any extra information needed. If there are gaps, the Web Search Agent conducts focused searches to find additional data, while Sub-Analysts provide more in-depth analysis.
- The Writing Agent combines all the information collected, including insights from specialized Sub-Analysts, into a complete market commentary. Finally, the Verification Agent checks the commentary for accuracy and clarity.

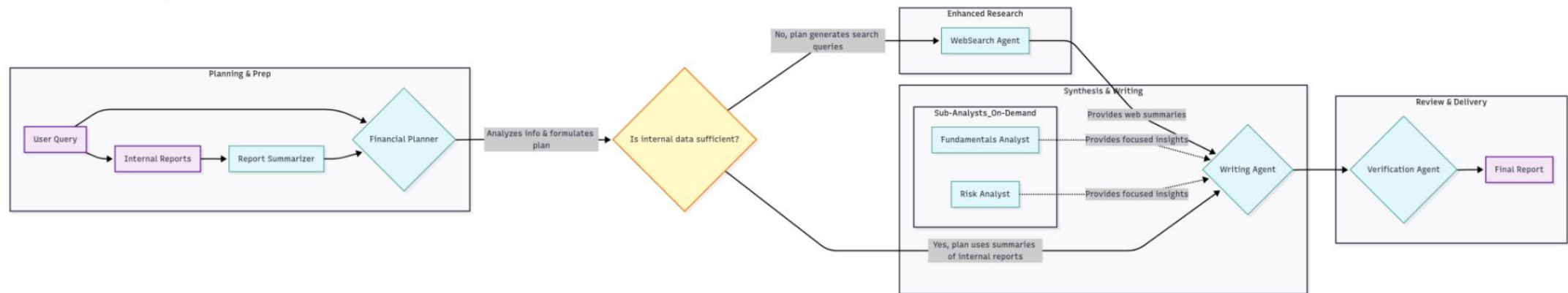


Figure 1. Flowchart of Collaborative Agents for Market Commentary

The Market Is a Network, Not a List

How Graph Neural Networks Are Capturing the True Topology of Finance
for Superior Alpha and Risk Management

Modeling Assets in Isolation.

Classical Econometrics

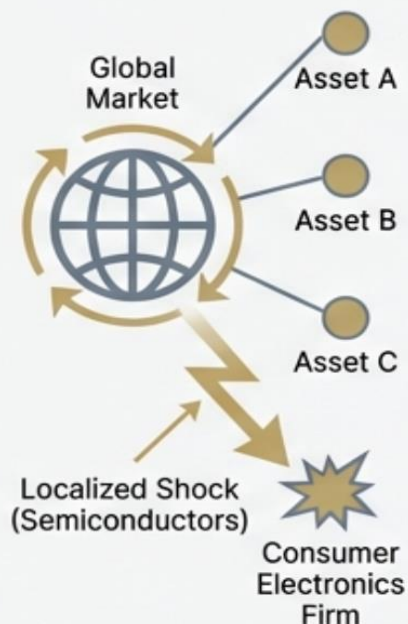
e.g., CAPM, Fama-French

Concept

Relies on linear relationships and global factors.

Fails to capture idiosyncratic, non-linear interactions and localized shocks.

Example: A supply chain disruption in semiconductors having a direct, non-linear impact on downstream consumer electronics firms that global factors miss.



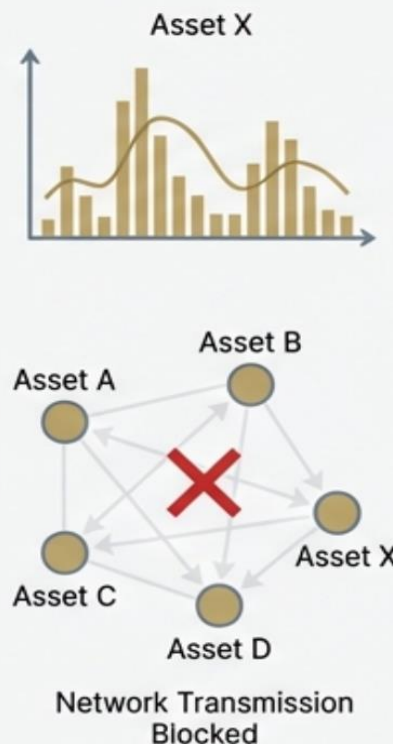
Univariate Volatility Models

e.g., GARCH

Concept

Models an asset's variance based on its own past shocks, capturing volatility clustering.

Cannot model the complex, multi-dimensional transmission of volatility across a network, especially during crises when correlations converge.



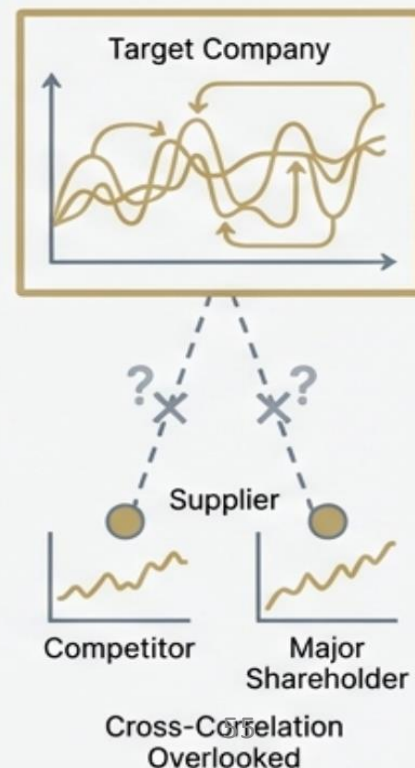
Temporal Deep Learning

e.g., LSTMs/GRUs

Concept

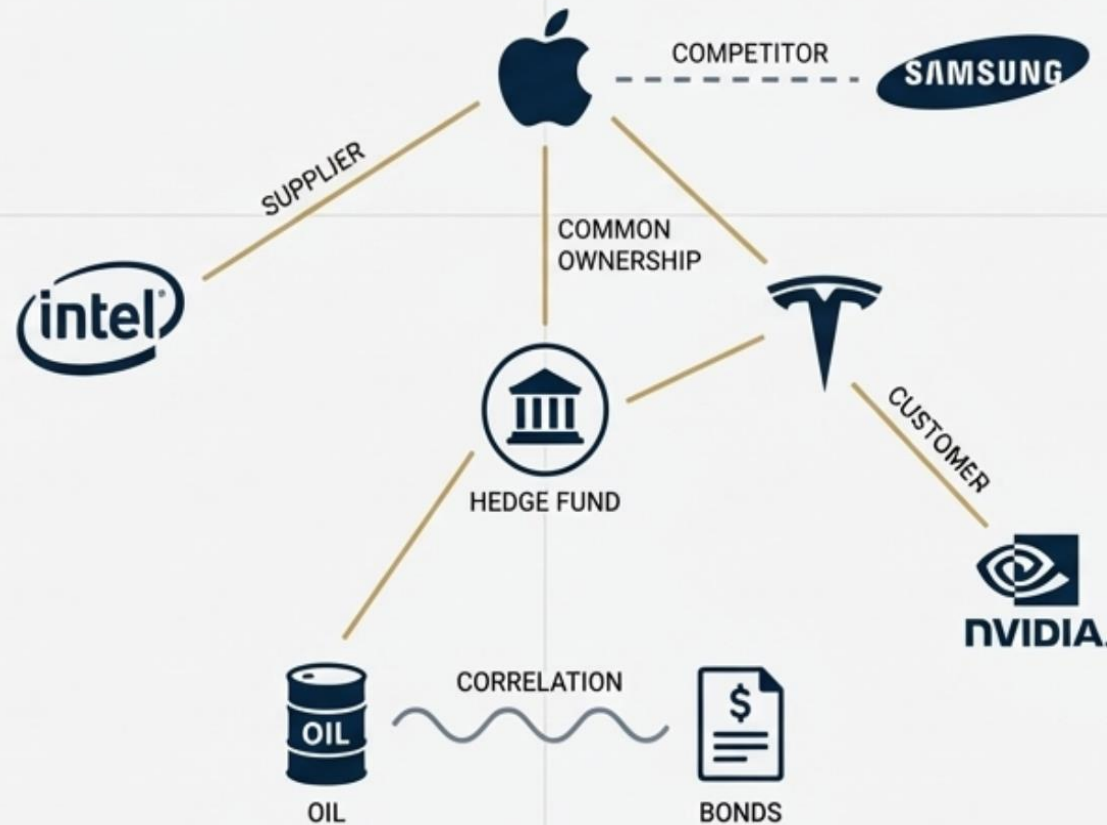
Revolutionized time series forecasting by capturing long-term temporal dependencies.

Suffers from “**spatial blindness**.” It processes a company's history in isolation, making it suboptimal at capturing cross-correlation between assets like suppliers, competitors, or major shareholders.



A New Paradigm: The “Market as a Graph” Hypothesis

Financial data is best represented not as a collection of independent vectors, but as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where **nodes** ($\mathcal{V}$) are financial entities and edges ($\mathcal{E}$) represent their relationships.



Spillover Effects

Financial momentum and distress are not contained; they propagate through the network. GNNs model this propagation via “message passing,” where a node’s state is updated by its neighbors.

Dynamic & Heterogeneous

The graph is not static. Relationships (edges) change over time, and there are multiple types of nodes and edges. This complexity requires advanced architectures.

Advanced Gen AI for Financial Time Series

- **Autoregressive Models**

- The GPT family of models uses a Transformer architecture and is typically trained as autoregressive models for sequence tasks.
 - TimeGPT (Nixtla) / TimesFM (Google) / Chronos (Amazon) excel in finance forecasting by utilizing a large-scale foundation model pre-trained on diverse time series to perform zero-shot or few-shot* prediction across various financial instruments with high accuracy.

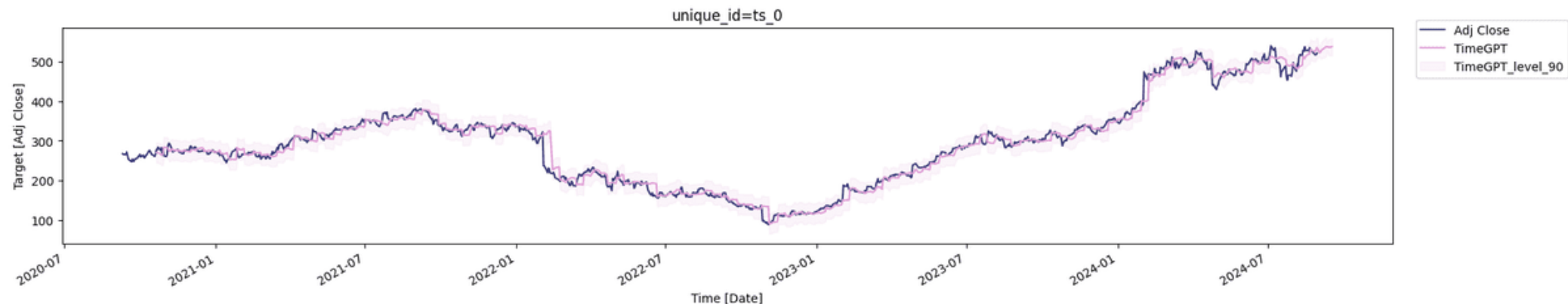
- **Generative models**

- Generative models primarily focus on learning the underlying probability distribution of data and generating new samples from that distribution.
 - Variational Autoencoder (VAE) captures the underlying probabilistic structure of financial data by encoding it into a compressed latent space, which is then used for anomaly detection, representation learning, and generating diverse synthetic market scenarios.
 - Generative Adversarial Network (GAN) is applied to use an adversarial training process between a generator and discriminator to create highly realistic and statistically accurate synthetic time series data, which is vital for stress- testing and privacy-preserving simulations.

Autoregressive Model - TimeGPT

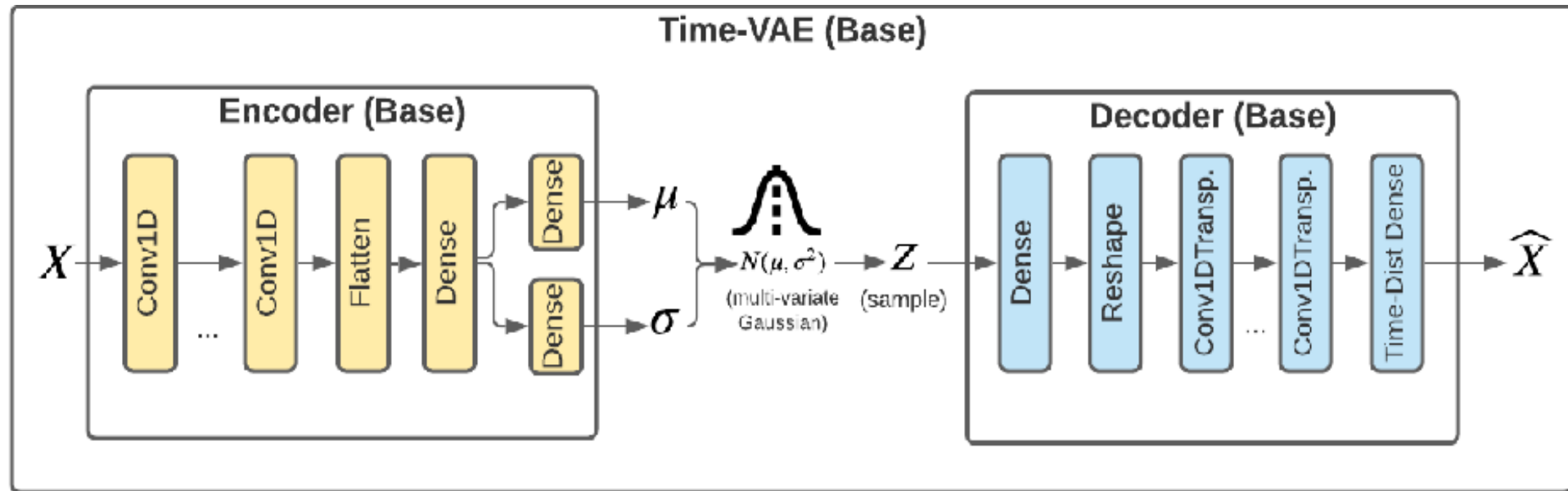
- TimeGPT is a large-scale, pre-trained model (specifically, a **Transformer-based** architecture with **self-attention**) trained on an enormous and highly diverse dataset of time series data.

Feature	Covariance Matrix (majorly used in Statistical Model)	Attention Mechanism (Self-Attention)
Primary Goal	Statically measure the linear relationship (variance and co-variance) between pairs of variables (features) in a dataset.	Dynamically weigh and aggregate information from an input sequence to enhance the representation of a specific element (Input Query).
Context	A statistical tool used for data analysis and dimensionality reduction	A component within a deep learning neural network (e.g., Transformer).



Generative Model - VAE

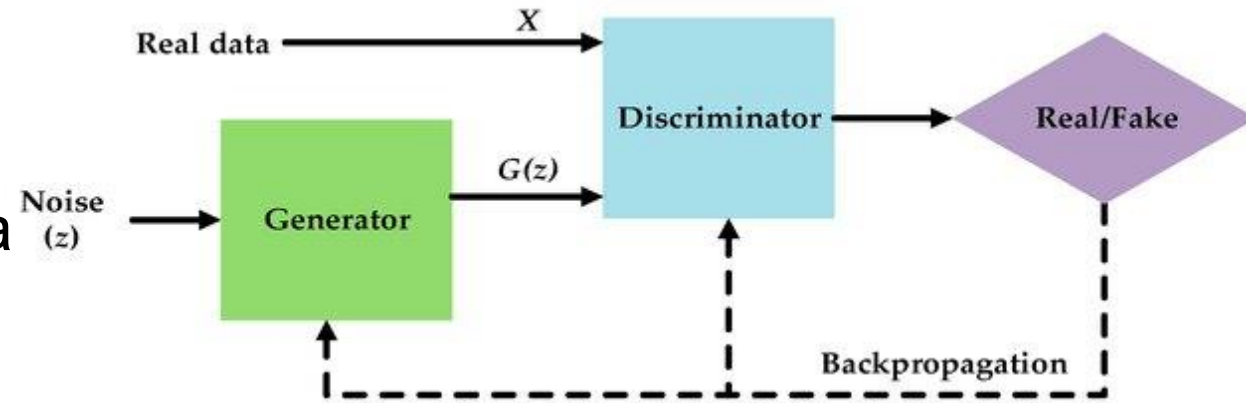
- The key idea of VAE is to learn a compressed, meaningful representation of the series that can be used to generate new, realistic time series or help in tasks like anomaly detection.



- The Encoder crushes that long, complex sequence into a tiny, compressed summary called **latent representation**
- The compressed summary becomes a **probabilistic distribution** in the latent space - **smooth and continuous**.
- The Decoder uncompresses the summary and tries to **reconstruct the original time series** as accurately as possible

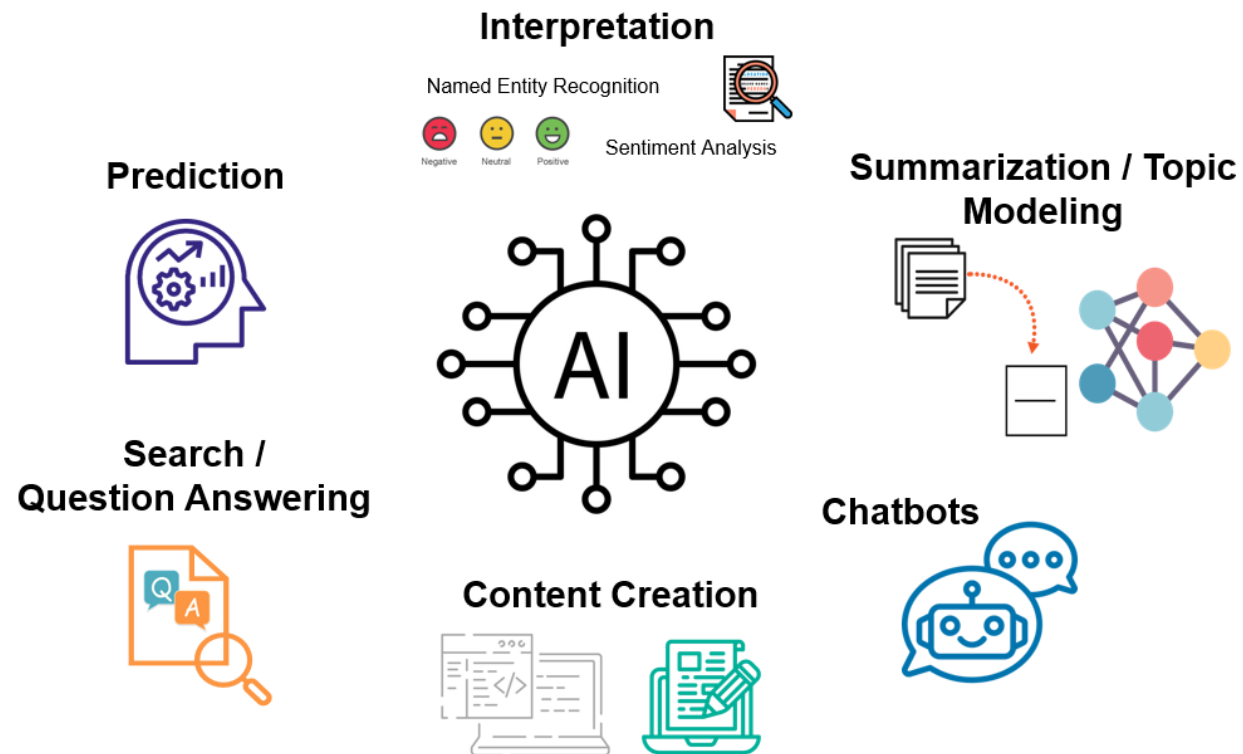
Generative Model - GAN

- The key idea of GAN model for time series data is to leverage the core adversarial training process to generate synthetic time series sequences that are indistinguishable from real-world data.
- While a standard GAN focuses only on "does this look real?", TimeGAN adds a second requirement: "does this follow the correct sequence?"
- Standard GAN: Might generate a stock path that looks like a stock chart visually, but statistically violates temporal rules (e.g., autocorrelation is broken).
- TimeGAN: Captures the temporal dynamics of sequences and ensures that if the original data had a high correlation between yesterday's volume and today's volatility, the synthetic data preserves that exact mathematical relationship.



Conclusion

- AI/LLMs have broad applications within asset management
- While these models will need care, competitive advantages can be created through customizations, training, fine-tuning and expert prompts
- “Iron-Person” can be more effective and efficient



Part 3:

AI Impact to Employment

How AI is Transforming Work at Anthropic

1. Anthropic engineers and researchers use Claude most often for fixing code errors and learning about the codebase.
2. People report increasing Claude usage and productivity gains.
3. **27%** of Claude-assisted work consists of tasks that wouldn't have been done otherwise, such as scaling projects, making nice-to-have tools
4. Most employees use Claude frequently while reporting they can “fully delegate” **0-20%** of their work to it.

Six Facts about the Recent Employment Effects of Artificial Intelligence

- This paper* examines changes in the labor market for occupations exposed to GenAI using high-frequency administrative data from the largest payroll software provider in the United States.
- This study uses data from ADP, the largest payroll processing firm in America. The company provides payroll services for firms employing over 25 million workers in the US.

Fact 1: Employment for Young Workers Has Declined in AI-exposed Occupations

- Employment for the youngest workers declines considerably after 2022, while employment for other age groups continues to grow.

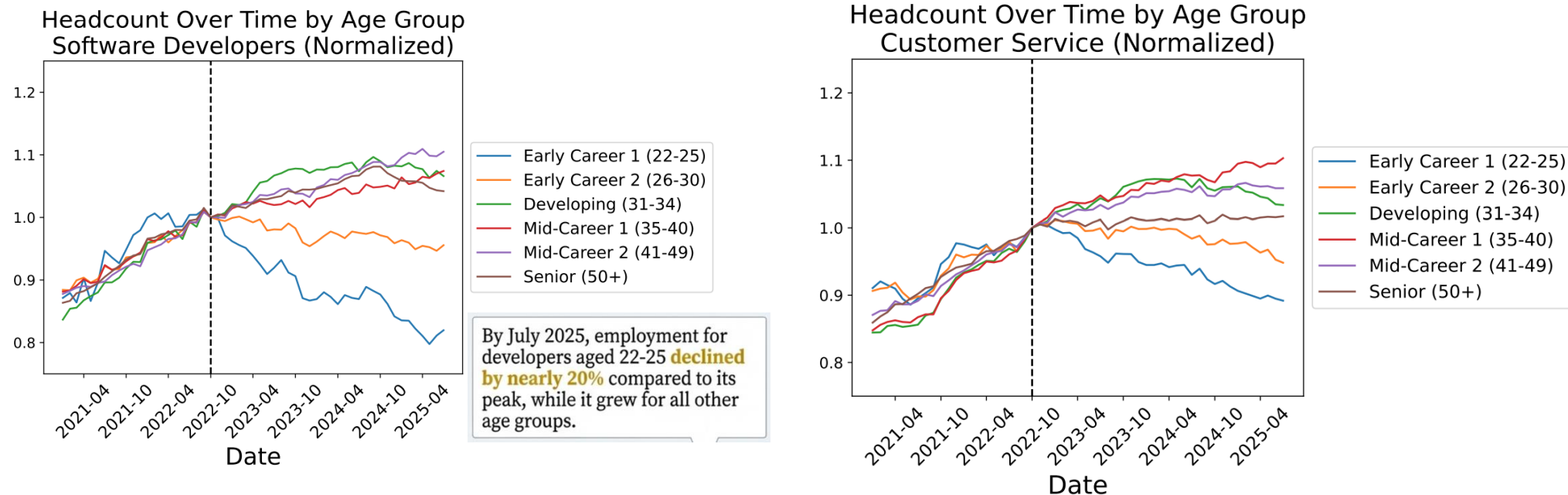
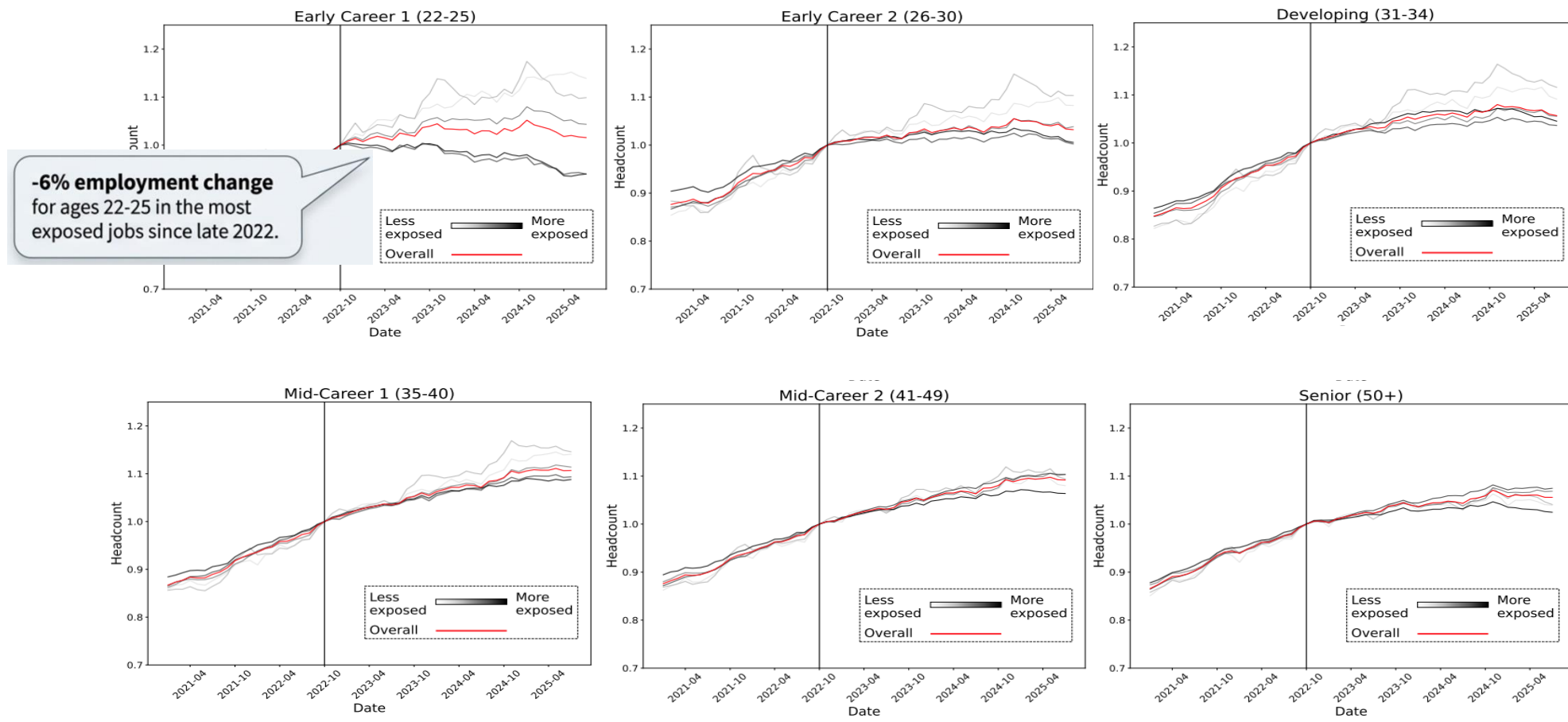


Figure 1: Employment changes for software developers and customer service agents by age, normalized to 1 in October 2022.

- There is a divergence in employment outcomes for more and less exposed occupations for workers aged 22-25, with more exposed occupations experiencing declining employment.
- For older age groups, it is found much less marked differences in employment growth across AI exposure quintiles



Fact 2: Though overall employment continues to grow, employment growth for young workers in particular has been stagnant

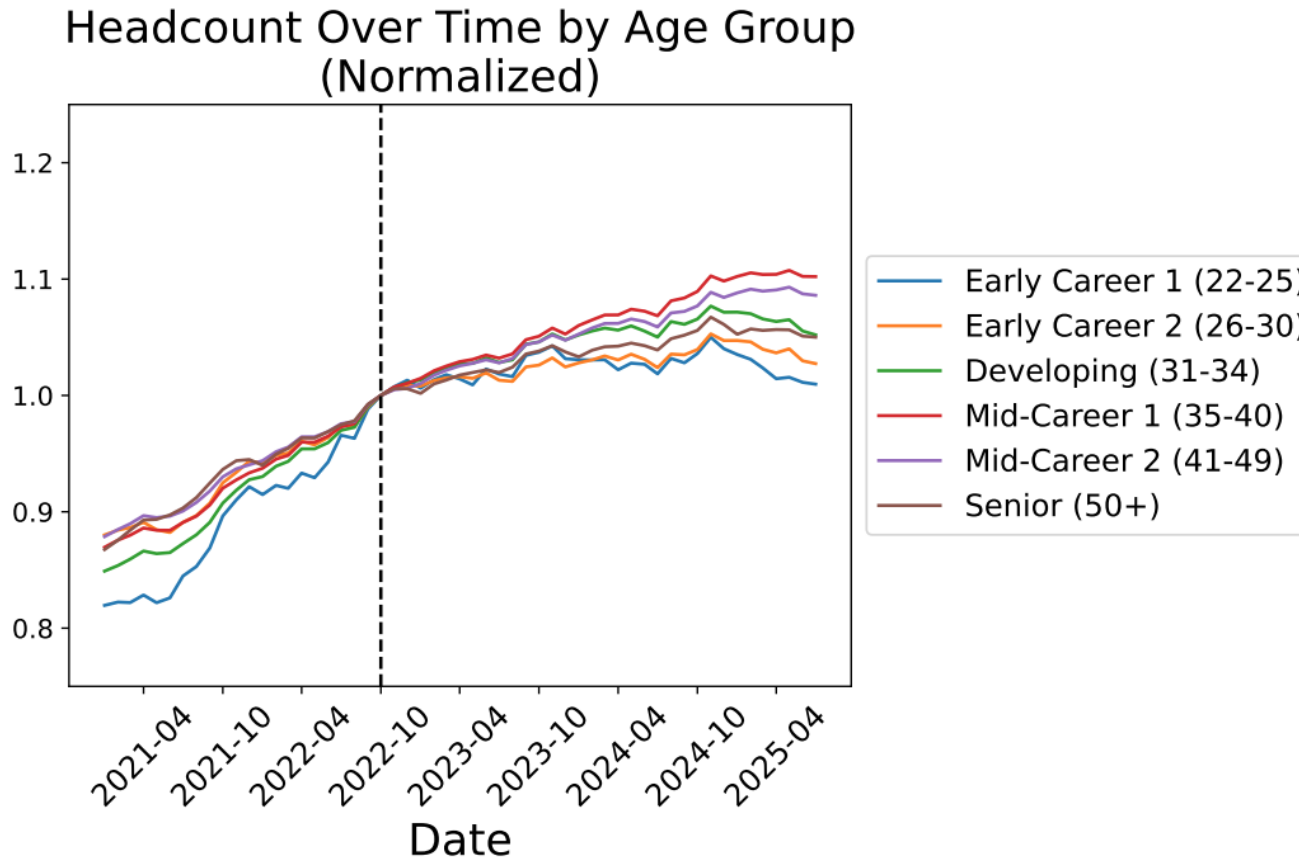
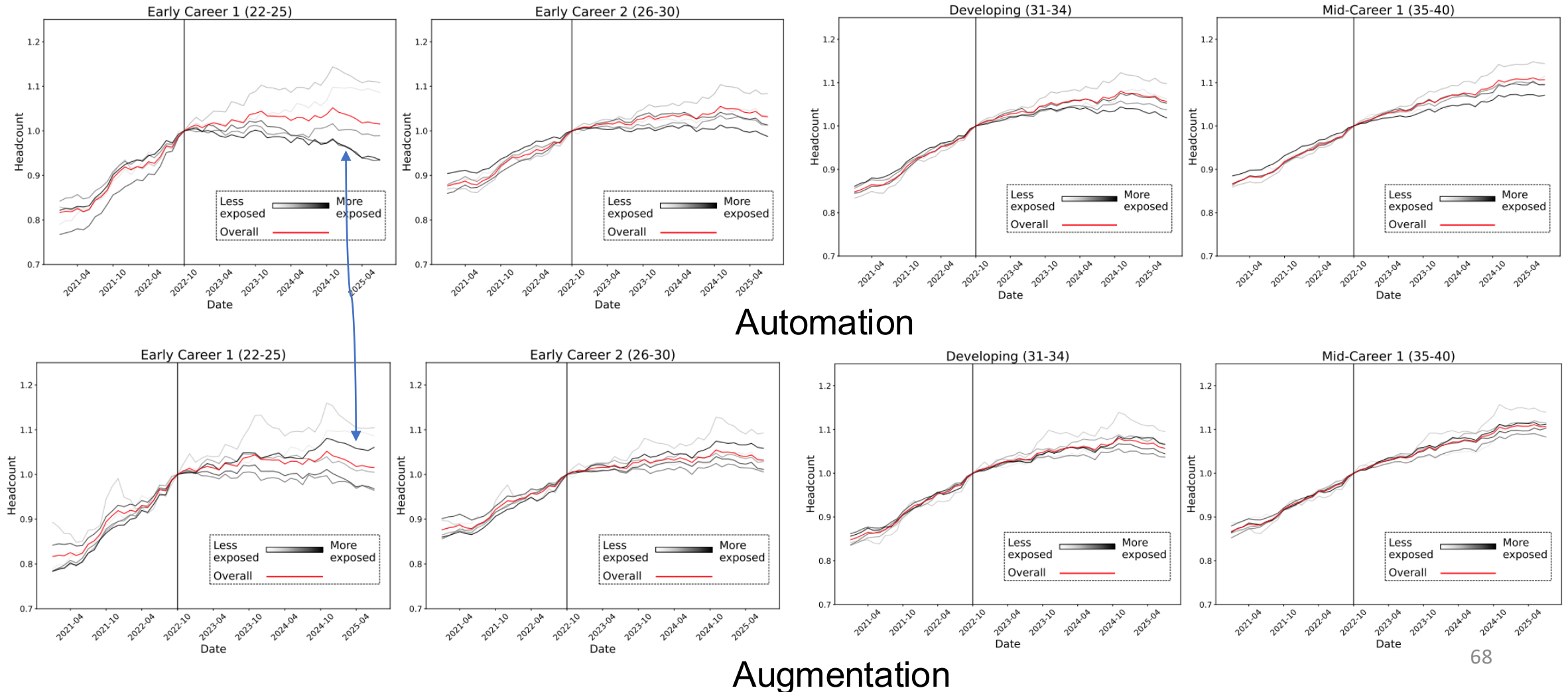


Figure 4: Employment changes by age. Including all occupations.

Fact 3: Entry-level employment has declined in applications of AI that automate work, with muted changes for augmentation



The Signal is Unmistakable and Robust

The observed employment decline for young workers is not an illusion caused by other economic factors. The findings hold even after controlling for firm-specific shocks, and the adjustments are appearing in headcount, not wages.



Fact 4: Not a Firm-Level Shock

After controlling for shocks affecting entire firms (like interest rate changes), a **13% relative employment decline** persists for young workers in the most exposed jobs versus the least.



Fact 5: It's About Jobs, Not Pay

Compensation trends remain flat across all groups. The labor market is adjusting through employment, not wages, suggesting wage stickiness.



Fact 6: Not Just a 'Tech' or 'Remote Work' Story

The patterns hold true even when excluding all computer occupations and firms in the information sector. They also appear in non-teleworkable jobs.

This is a real, AI-driven labor market effect.

Key Findings

1. Substantial declines are observed in employment for early-career workers in occupations most exposed to AI, e.g., software development & customer support.
2. Authors show that economy-wide employment continues to grow, but employment growth for young workers has been stagnant.
3. Entry-level employment has declined in applications of AI that automate work, with muted effects for those that augment it.
4. These employment declines remain after conditioning on firm-time effects, with a 13% relative employment decline for young workers in most exposed occupations.
5. These labor market adjustments are more visible in employment than in compensation.
6. Authors find that these patterns hold in occupations unaffected by remote work and across various alternative sample constructions.

We Are Measuring a 21st Century Shift with 20th Century Tools

Traditional workforce metrics were designed for a physical economy where jobs are tied to geographic locations. They are blind to the emerging 'AI Economy' of AI-mediated work, Human-AI teams, and digital value creation that happens on private platforms.

Physical Economy (Census Visible)



- Manufacturing Jobs



- Service Employment



- Worker Residences

Physical economy = measurable economy. People work in places, Census counts people and places.

AI Economy (A Blind Spot)



- AI-mediated work



- Human-AI teams



- AI Coordination

AI economy ≠ measurable economy. AI automation tools have no state ID; they work through private platforms the census can't see.

Introducing the Iceberg Index: A New Lens for the AI Economy

The Iceberg Index is a skills-centered metric that measures the **wage value of skills** that current AI systems can technically perform within each occupation.



- **It Measures: Technical Exposure**

Where human and AI capabilities overlap today, based on a catalog of over 13,000 production AI tools.

- **It Does NOT Predict: Job Losses or Adoption Timelines**

It is a capability map, like an earthquake risk zone, that enables proactive planning, not a deterministic forecast.



1. Capture

Map 151M+ human workers & 32,000+ skills.



2. Analyze

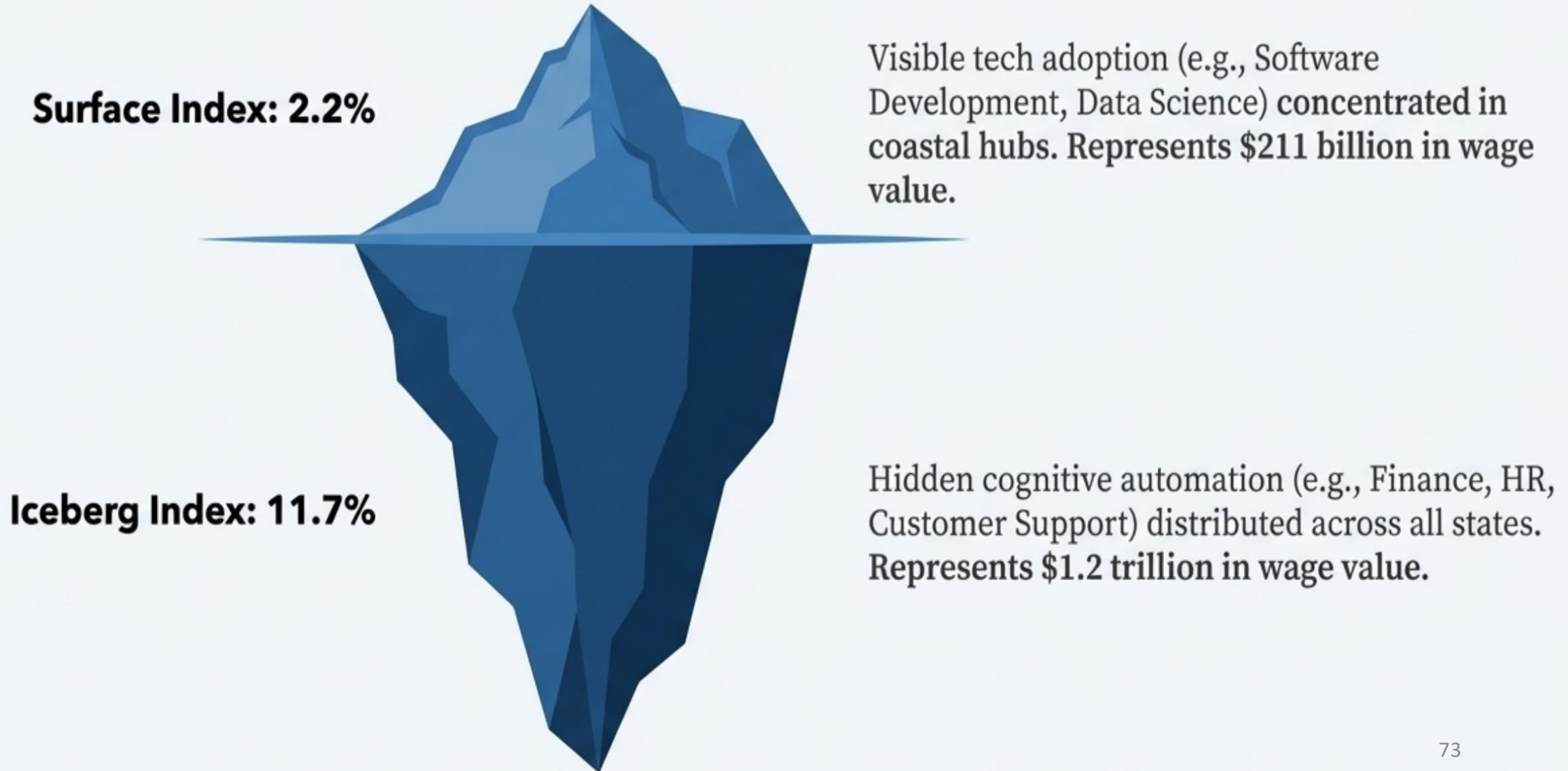
Measure 13,000+ AI tools against the same skills.



3. Simulate

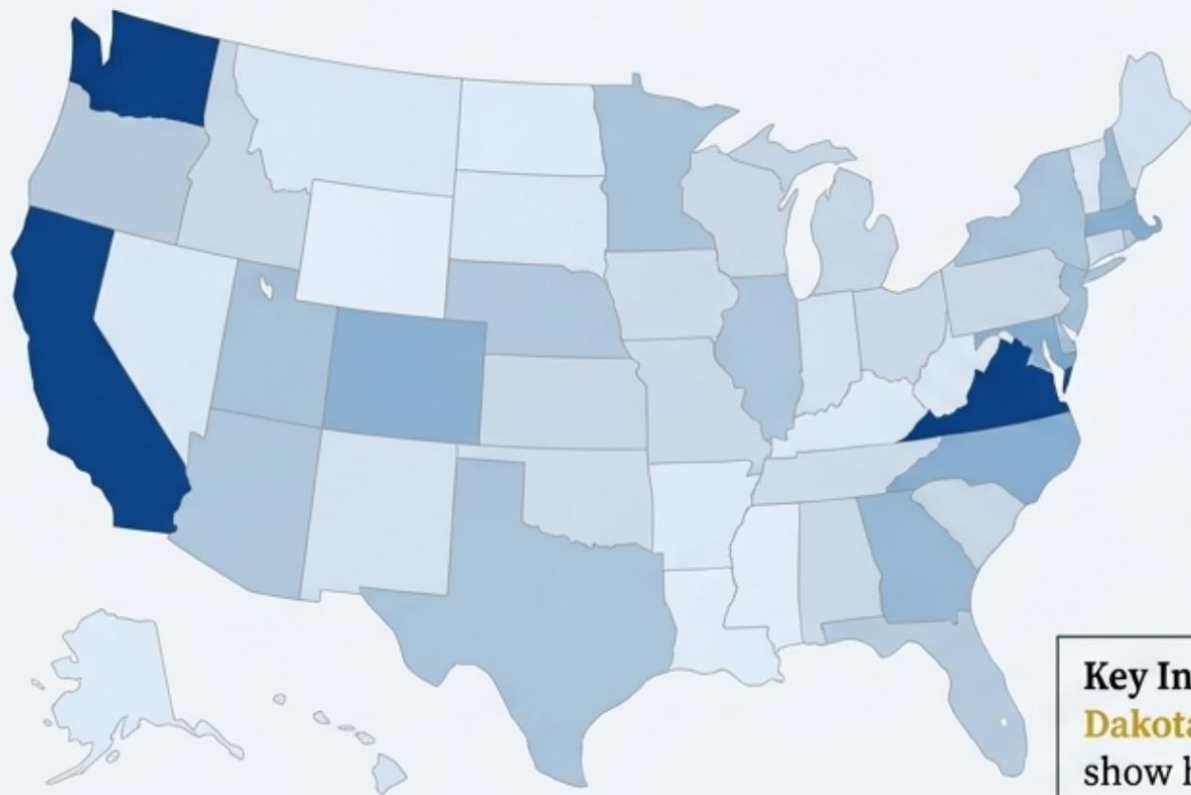
Model human-AI interaction to calculate exposure.

The Hidden Exposure is Five Times Larger Than What We See

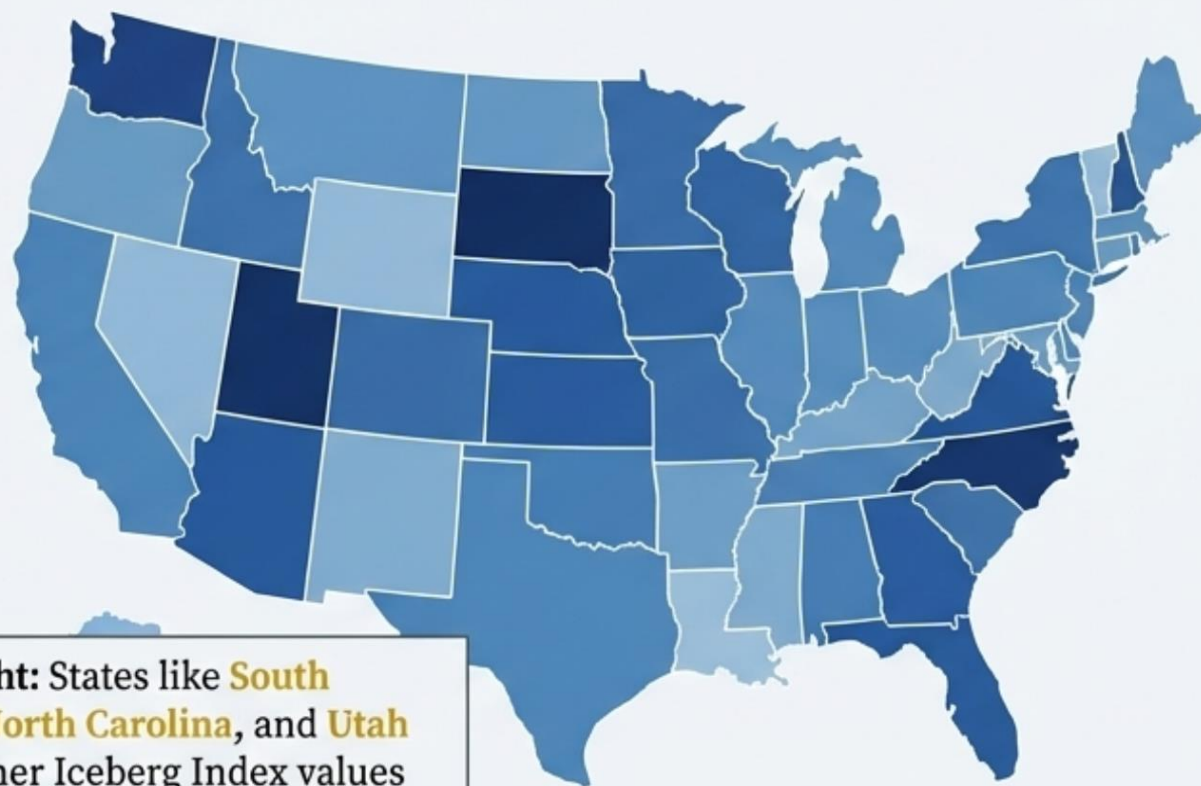


AI Exposure Is Not a Coastal Story. It Is an American Story.

Visible Tech Adoption (Surface Index)



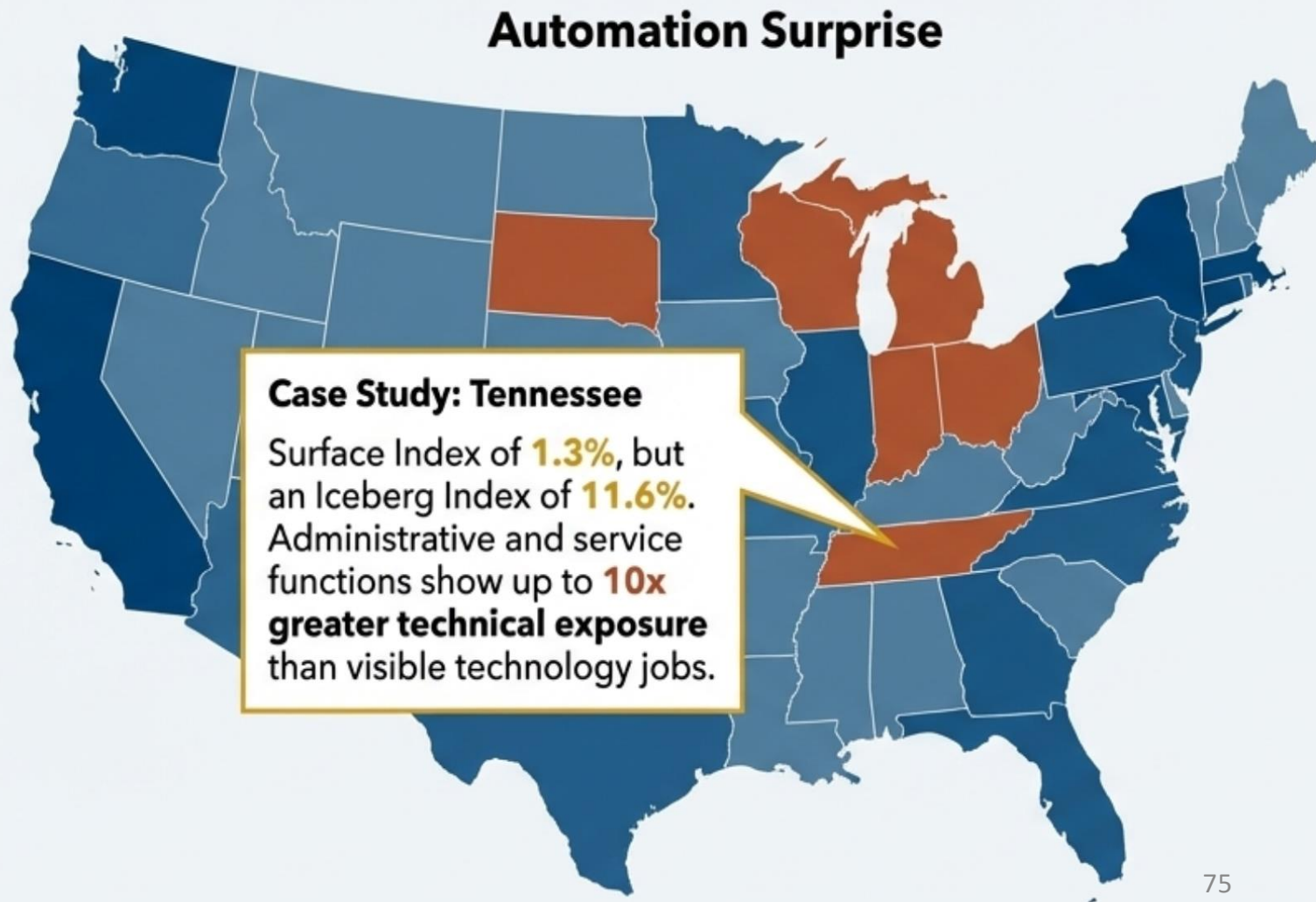
Hidden Nationwide Exposure (Iceberg Index)



Key Insight: States like **South Dakota**, **North Carolina**, and **Utah** show higher Iceberg Index values than California or Virginia, driven by their concentration of administrative, financial, and professional service roles.

The Blind Spot: Manufacturing States Face a Hidden White-Collar Surprise

The largest gaps between visible tech exposure and hidden cognitive exposure are in America's industrial heartland. While these states may be focused on physical automation, the more immediate exposure is in the administrative, financial, financial, and coordination roles that support their core industries.



Key Takeaways

Project Iceberg

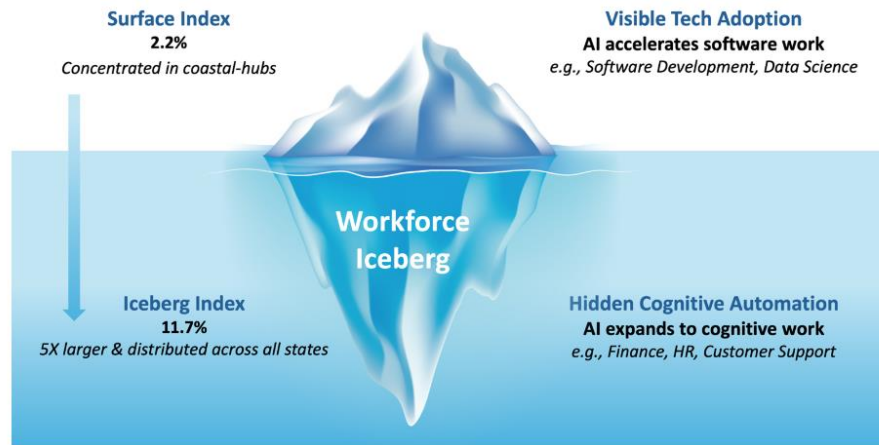


Figure 3. The Iceberg Index reveals workforce exposure five times larger than visible tech adoption. Above the waterline: the Surface Index (2.2%) tracks current AI adoption in coastal technology hubs. Below the waterline: the Iceberg Index (11.7%) measures technical capability spanning administrative, financial, and professional services nationwide. This hidden mass represents where workforce preparation strategies based solely on visible tech-sector signals may fall short.

strategies, and allocate resources where they will have the most impact.

Iceberg Simulations: Scale and Infrastructure

- **Workforce Scale:** 151 million workers across 923 occupations in 3,000 counties
- **Skill Coverage:** 32,000+ distinct skills from standardized taxonomies
- **AI Systems:** 13,000+ tools across software, cognitive, and physical capabilities
- **Simulation Engine:** Large Population Models used for national security [10].
- **Computing Platform:** Frontier supercomputer at Oak Ridge National Lab [21].

- **Iceberg Index**, a metric that measures the wage value of skills AI can technically perform.
 - **The Surface (Visible):** AI adoption in computing and technology represents only **2.2%** of wage value (approx. \$211 billion)
 - **The Submerged (Hidden):** The technical capability for AI to automate cognitive, administrative, and financial tasks represents **11.7%** of wage value (approx. \$1.2 trillion)
- **White-Collar Exposure is Nationwide** While tech disruption is concentrated in specific hubs, the "hidden mass" of exposure is distributed across the entire country.
- **The "Automation Surprise" in Manufacturing States**
 - Policymakers in these regions often focus on physical automation (robotics). However, some states face massive, immediate exposure in **white-collar support roles**—such as logistics coordination, supply chain finance, and administrative oversight.
 - Tennessee, for example, has a low "Surface Index" (tech jobs) but a very high "Iceberg Index" due to these support functions.